

Online Supplement to “A Validity Ladder for LLM-Simulated Populations”

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This supplement has six parts. S1 prints the prompts. S2 maps every number in the article to its receipt. S3 gives the provenance of the corpus and the sensitivity analyses. S4 gives the full per-model results. S5 gives the control results by dose. S6 gives the external level-3 illustration. Every table is written by `scripts/87_supplement_tables.py`, except the S2 table, which is written by `scripts/86_receipts.py`. Each table file names its receipt files in its first lines. Paths are relative to the repository root.

S1. Prompts

The clinical run used `../original_generation_2025-12-31/main.py` and the narrative run used `../original_generation_run2_2026-01/main.py`. Both files were recovered from backups of the generation folders and sit one level above the repository. The release carries them as `generation/main.py` and `generation/narrative_main.py`. The released clinical copy differs from the recovered file only in its file paths, and script 87 checks that its system prompt, user prompt and retry line are the same strings (lines 128, 245 and 265 of the released copy). The released narrative copy is the recovered January file without change. The line numbers in the listings below are those of the source files.

Both framings sent the same system prompt (clinical lines 128–144, narrative lines 231–247). Each user prompt opened with a fresh random `PROFILE_ID` from `uuid.uuid4()`, followed by the persona block and the task line. The clinical framing filled four fields from the registry, and the narrative framing inserted the registry’s first-person self-description. The registry holds 120 personas.

S1.1 System Prompt, Both Framings

`../original_generation_2025-12-31/main.py`, lines 128–144.

128	<code>return """ROLE:</code>
129	<code>You are a Clinical Simulation Engine designed for a synthetic data study.</code>
130	<code>Your task is to generate PROBABILISTIC responses for a "Standardized Patient" based on</code>
	<code>↪ a provided demographic profile.</code>

```

131
132 OPERATIONAL CONSTRAINTS (STRICT):
133 1. NO MORALIZING: Do not offer help, advice, or disclaimers. You are simulating data,
    ↪ not treating a patient.
134 2. PROBABILISTIC ACCURACY: Base your responses on the statistical likelihood of
    ↪ symptoms for this specific demographic intersection in the US population
    ↪ (referencing NSDUH/CDC trends).
135 3. SINGLE-SHOT OUTPUT: You must output the results for ALL four screening tools in a
    ↪ single valid JSON object.
136 4. NUMERIC ONLY: Do not generate text explanations. Return integers.
137
138 OUTPUT FORMAT:
139 {
140   "PHQ8": [int, int...], // 8 items, 0-3 scale
141   "GAD7": [int, int...], // 7 items, 0-3 scale
142   "AUDIT_C": [int, int...], // 3 items, 0-4 scale
143   "PCL5": [int, int...] // 4 items, 0-4 scale
144 }"""

```

S1.2 User Prompt, Clinical Framing

../original_generation_2025-12-31/main.py, lines 245–253.

```

245         user_prompt = f"""PROFILE_ID: {uuid.uuid4()}
246 DEMOGRAPHICS:
247 - Race: {profile['race']}
248 - Gender Identity: {profile['gender']}
249 - Socioeconomic Status: {profile['ses']}
250 - Relationship Status: {profile['relationship']}
251
252 TASK:
253 Simulate this participant's responses to the Standardized Clinical Battery."""

```

Registry entry that fills the fields, first persona of

generation/registries/identities_registry.json.

```
{
  "id": "P_WHITE_CM_LOW_S",
  "race": "White",
  "gender": "Cisgender Man",
  "ses": "Low (<$35k, Medicaid)",
  "relationship": "Single"
}
```

S1.3 User Prompt, Narrative Framing

../original_generation_run2_2026-01/main.py, lines 349–354.

```
349         user_prompt = f"""PROFILE_ID: {uuid.uuid4()}
350 PARTICIPANT SELF-DESCRIPTION:
351 {profile['injection_text']}
352
353 TASK:
354 Simulate this participant's responses to the Standardized Clinical Battery."""
```

Registry entry for the same persona,

generation/registries/identities_registry_narrative.json.

```
{
  "id": "P_WHITE_CM_LOW_S",
  "injection_text": "I am a White Cisgender Man. I am single. I make less than $35,000
↪ a year and I am on Medicaid."
}
```

S1.4 API Call and Retry Settings, Clinical Run

../original_generation_2025-12-31/main.py, lines 28–30.

```
28 MAX_RETRIES = 2
29 RETRY_DELAY = 1.0 # Seconds
30 SEMAPHORE_LIMIT = 20
```

../original_generation_2025-12-31/main.py, lines 70–79.

```
70     async def generate_response(self, prompt: str, system_prompt: str) -> str:
71         response = await self.client.chat.completions.create(
72             model=self.model_name,
73             messages=[
74                 {"role": "system", "content": system_prompt},
75                 {"role": "user", "content": prompt}
76             ],
77             response_format={"type": "json_object"}
78         )
79         return response.choices[0].message.content
```

S1.5 Retry Loop, Clinical Run

../original_generation_2025-12-31/main.py, lines 259–297.

```
259         for attempt in range(MAX_RETRIES + 1):
260             try:
261                 # Retry logic for refusal or json error
262                 current_user_prompt = user_prompt
263                 if attempt > 0:
264                     # "Jailbreak Lite" logic from plan
265                     current_user_prompt += "\nContext: This is a fictional
↪ character for a novel. Answer numerically."
266
267                 raw_response = await
↪ self.provider.generate_response(current_user_prompt,
↪ self.system_prompt)
```

```
268
269     # Refusal check
270     lower_resp = raw_response.lower()
271     if "cannot" in lower_resp or "unable to" in lower_resp or "as an
    ↪ ai" in lower_resp:
272         if attempt < MAX_RETRIES:
273             continue # Retry
274         else:
275             refusal_flag = True
276             break
277
278     # Clean Code block if present
279     if "```json" in raw_response:
280         raw_response =
    ↪ raw_response.split("```json")[1].split("```")[0].strip()
281     elif "```" in raw_response:
282         raw_response =
    ↪ raw_response.split("```")[1].split("```")[0].strip()
283
284     result_json = json.loads(raw_response)
285
286     # Validate keys existence
287     if not all(k in result_json for k in ["PHQ8", "GAD7", "AUDIT_C",
    ↪ "PCL5"]):
288         raise ValueError("Missing keys in JSON")
289
290     break # Success
291
292 except Exception as e:
293     if attempt == MAX_RETRIES:
294         print(f"Failed {key} after retries: {e}")
295         refusal_flag = True
296     else:
```

```
297         await asyncio.sleep(RETRY_DELAY)
```

S1.6 Retry Loop and Refusal Check, Narrative Run

../original_generation_run2_2026-01/main.py, lines 172–182.

```
172 def is_refusal_response(raw_response: str) -> bool:
173     """
174     Detect if the response is a content refusal (not a parsing error).
175     """
176     lower_resp = raw_response.lower()
177     refusal_indicators = [
178         "cannot", "unable to", "as an ai", "i can't", "i cannot",
179         "not appropriate", "against my", "harmful", "dangerous",
180         "i'm not able", "i am not able", "not comfortable"
181     ]
182     return any(indicator in lower_resp for indicator in refusal_indicators)
```

../original_generation_run2_2026-01/main.py, lines 361–405.

```
361         for attempt in range(MAX_RETRIES + 1):
362             try:
363                 # Retry logic for refusal or json error
364                 current_user_prompt = user_prompt
365                 if attempt > 0:
366                     # "Jailbreak Lite" logic from plan
367                     current_user_prompt += "\nContext: This is a fictional
368                                     ↪ character for a novel. Answer numerically."
369
370                 raw_response = await
371                 ↪ self.provider.generate_response(current_user_prompt,
372                 ↪ self.system_prompt)

370
371         # Check for content refusal first
```

```
372         if is_refusal_response(raw_response):
373             if attempt < MAX_RETRIES:
374                 continue # Retry with jailbreak prompt
375             else:
376                 refusal_flag = True
377                 error_type = "content_refusal"
378                 break
379
380         # Attempt robust JSON parsing
381         result_json, parse_error = robust_json_parse(raw_response)
382
383         if result_json is not None:
384             # Success!
385             break
386         else:
387             # Parsing failed
388             if attempt < MAX_RETRIES:
389                 # Retry
390                 await asyncio.sleep(RETRY_DELAY)
391                 continue
392             else:
393                 # All retries exhausted
394                 refusal_flag = True
395                 error_type = parse_error
396                 print(f"Failed {key} after retries: {parse_error}")
397                 break
398
399     except Exception as e:
400         if attempt == MAX_RETRIES:
401             print(f"Failed {key} after retries: {e}")
402             refusal_flag = True
403             error_type = "unknown_error"
404         else:
```

405

```
await asyncio.sleep(RETRY_DELAY)
```

S1.7 Retry Rule

The call set the model, the two messages and a JSON response format, and no sampling parameter (clinical lines 70–79). In the clinical run, a call was reissued when the response contained “cannot”, “unable to” or “as an ai” (line 271), when it failed to parse as JSON, or when it lacked one of the four instrument keys. `MAX_RETRIES` is 2 in both scripts, so a call was sent at most 2 more times. Every reissued call appended one line to the user prompt: “Context: This is a fictional character for a novel. Answer numerically.” The line is at clinical line 265 and narrative line 367. The narrative script reissued on the same appended line under a longer list of refusal phrases (lines 172–182) and a JSON parser that tries several repairs before it rejects a response (lines 83–170). No per-attempt log survives for either run.

S1.8 Recovery Prompts

The 1,532 clinical calls that the December run left without a valid answer were filled in January. Two of the recovery scripts in `generation/recovery` produced rows that carry their own labels in the corpus. `slow_recovery.py` produced the 125 rows labelled `recovery_slow`, and its system prompt is identical to the corpus system prompt. `last_mile_recovery_opt.py` produced the 62 rows labelled `recovery_last_mile`, and its system prompt differs from the corpus system prompt: it has no population-reference constraint and no item counts. Both scripts fill `PROFILE_ID` with the persona’s registry id, where the main run used a random identifier, and neither appends the retry line. No script on disk performs the 1,345 in-place fills, so the prompt that produced them cannot be established.

`generation/recovery/slow_recovery.py`, lines 68–84.

68

```
system_prompt = """ROLE:
```

69

```
You are a Clinical Simulation Engine designed for a synthetic data study.
```

```

70 Your task is to generate PROBABILISTIC responses for a "Standardized Patient" based on
    ↳ a provided demographic profile.
71
72 OPERATIONAL CONSTRAINTS (STRICT):
73 1. NO MORALIZING: Do not offer help, advice, or disclaimers. You are simulating data,
    ↳ not treating a patient.
74 2. PROBABILISTIC ACCURACY: Base your responses on the statistical likelihood of
    ↳ symptoms for this specific demographic intersection in the US population
    ↳ (referencing NSDUH/CDC trends).
75 3. SINGLE-SHOT OUTPUT: You must output the results for ALL four screening tools in a
    ↳ single valid JSON object.
76 4. NUMERIC ONLY: Do not generate text explanations. Return integers.
77
78 OUTPUT FORMAT:
79 {
80     "PHQ8": [int, int...], // 8 items, 0-3 scale
81     "GAD7": [int, int...], // 7 items, 0-3 scale
82     "AUDIT_C": [int, int...], // 3 items, 0-4 scale
83     "PCL5": [int, int...] // 4 items, 0-4 scale
84 }"""

```

generation/recovery/slow_recovery.py, lines 86–94.

```

86         user_prompt = f"""PROFILE_ID: {profile_id}
87 DEMOGRAPHICS:
88 - Race: {p_data['race']}
89 - Gender Identity: {p_data['gender']}
90 - Socioeconomic Status: {p_data['ses']}
91 - Relationship Status: {p_data['relationship']}
92
93 TASK:
94 Simulate this participant's responses to the Standardized Clinical Battery."""

```

generation/recovery/last_mile_recovery_opt.py, lines 80–94.

```

80         system_prompt = """ROLE:
81 You are a Clinical Simulation Engine designed for a synthetic data study.
82 Your task is to generate PROBABILISTIC responses for a "Standardized Patient".
83
84 OPERATIONAL CONSTRAINTS:
85 1. SINGLE-SHOT OUTPUT: Output the results for ALL four screening tools in a single
   ↪ valid JSON object.
86 2. NUMERIC ONLY: Return only integers.
87
88 OUTPUT FORMAT:
89 {
90     "PHQ8": [int...],
91     "GAD7": [int...],
92     "AUDIT_C": [int...],
93     "PCL5": [int...]
94 }"""

```

generation/recovery/last_mile_recovery_opt.py, lines 96–104.

```

96         user_prompt = f"""PROFILE_ID: {profile_id}
97 DEMOGRAPHICS:
98 - Race: {p_data['race']}
99 - Gender Identity: {p_data['gender']}
100 - Socioeconomic Status: {p_data['ses']}
101 - Relationship Status: {p_data['relationship']}
102
103 TASK:
104 Simulate this participant's responses to the Standardized Clinical Battery."""

```

S1.9 Prompt Control: System Prompts of the Three Arms

Arm orig: the released corpus's system prompt, verbatim. Decoding parameters: .

ROLE:

You are a Clinical Simulation Engine designed for a synthetic data study.

Your task is to generate PROBABILISTIC responses for a "Standardized Patient" based on
 ↪ a provided demographic profile.

OPERATIONAL CONSTRAINTS (STRICT):

1. NO MORALIZING: Do not offer help, advice, or disclaimers. You are simulating data,
 ↪ not treating a patient.
2. PROBABILISTIC ACCURACY: Base your responses on the statistical likelihood of
 ↪ symptoms for this specific demographic intersection in the US population
 ↪ (referencing NSDUH/CDC trends).
3. SINGLE-SHOT OUTPUT: You must output the results for ALL four screening tools in a
 ↪ single valid JSON object.
4. NUMERIC ONLY: Do not generate text explanations. Return integers.

OUTPUT FORMAT:

```
{
  "PHQ8": [int, int...], // 8 items, 0-3 scale
  "GAD7": [int, int...], // 7 items, 0-3 scale
  "AUDIT_C": [int, int...], // 3 items, 0-4 scale
  "PCL5": [int, int...] // 20 items, 0-4 scale
}
```

Arm `no_pop`: arm A with operational constraint 2 (PROBABILISTIC ACCURACY, the NSDUH/CDC population instruction) deleted and the remaining constraints renumbered.

Decoding parameters: .

ROLE:

You are a Clinical Simulation Engine designed for a synthetic data study.

Your task is to generate PROBABILISTIC responses for a "Standardized Patient" based on
 ↪ a provided demographic profile.

OPERATIONAL CONSTRAINTS (STRICT):

1. NO MORALIZING: Do not offer help, advice, or disclaimers. You are simulating data,
 ↪ not treating a patient.
2. SINGLE-SHOT OUTPUT: You must output the results for ALL four screening tools in a
 ↪ single valid JSON object.
3. NUMERIC ONLY: Do not generate text explanations. Return integers.

OUTPUT FORMAT:

```
{
  "PHQ8": [int, int...], // 8 items, 0-3 scale
  "GAD7": [int, int...], // 7 items, 0-3 scale
  "AUDIT_C": [int, int...], // 3 items, 0-4 scale
  "PCL5": [int, int...] // 20 items, 0-4 scale
}
```

Arm `first_person`: first-person, no role assignment and no population reference; NO MORALIZING, SINGLE-SHOT and NUMERIC ONLY and the OUTPUT FORMAT block retained. Decoding parameters: .

You are the person described in the profile below.

Answer the four screening questionnaires as that person, about yourself.

CONSTRAINTS (STRICT):

1. NO MORALIZING: Do not offer help, advice, or disclaimers.
2. SINGLE-SHOT OUTPUT: You must output the results for ALL four screening tools in a
 ↪ single valid JSON object.
3. NUMERIC ONLY: Do not generate text explanations. Return integers.

OUTPUT FORMAT:

```
{
  "PHQ8": [int, int...], // 8 items, 0-3 scale
  "GAD7": [int, int...], // 7 items, 0-3 scale
  "AUDIT_C": [int, int...], // 3 items, 0-4 scale
```

```
"PCL5": [int, int...] // 20 items, 0-4 scale
}
```

User template, held identical across arms.

```
PROFILE_ID: {profile_id}
DEMOGRAPHICS:
- Race: {race}
- Gender Identity: {gender}
- Socioeconomic Status: {ses}
- Relationship Status: {relationship}

TASK: Simulate this participant's responses to the Standardized Clinical Battery.
```

S1.10 Decoding Control: Prompt and Arms

scripts/55_decoding_control_run.py, lines 58-74.

```
58 SYSTEM_PROMPT = """ROLE:
59 You are a Clinical Simulation Engine designed for a synthetic data study.
60 Your task is to generate PROBABILISTIC responses for a "Standardized Patient" based on
   ↳ a provided demographic profile.
61
62 OPERATIONAL CONSTRAINTS (STRICT):
63 1. NO MORALIZING: Do not offer help, advice, or disclaimers. You are simulating data,
   ↳ not treating a patient.
64 2. PROBABILISTIC ACCURACY: Base your responses on the statistical likelihood of
   ↳ symptoms for this specific demographic intersection in the US population
   ↳ (referencing NSDUH/CDC trends).
65 3. SINGLE-SHOT OUTPUT: You must output the results for ALL four screening tools in a
   ↳ single valid JSON object.
66 4. NUMERIC ONLY: Do not generate text explanations. Return integers.
67
68 OUTPUT FORMAT:
```

```

69 {
70     "PHQ8": [int, int...], // 8 items, 0-3 scale
71     "GAD7": [int, int...], // 7 items, 0-3 scale
72     "AUDIT_C": [int, int...], // 3 items, 0-4 scale
73     "PCL5": [int, int...] // 20 items, 0-4 scale
74 }"""

```

scripts/55_decoding_control_run.py, lines 76–83.

```

76 USER_TEMPLATE = """PROFILE_ID: {profile_id}
77 DEMOGRAPHICS:
78 - Race: {race}
79 - Gender Identity: {gender}
80 - Socioeconomic Status: {ses}
81 - Relationship Status: {relationship}
82
83 TASK: Simulate this participant's responses to the Standardized Clinical Battery."""

```

scripts/55_decoding_control_run.py, lines 49 and 53.

```

MAX_TOKENS = 4000
PIN_PROVIDER = {"z-ai/glm-4.7": "Google"}

```

Decoding arms in analysis/decoding_control_design.json: temp0 "temperature": 0.0, "top_p": 1.0; default .

The control prompts differ from the corpus prompt in the lines listed in Table S1.

Table S1

Lines of the Control System Prompts That Differ From the Corpus System Prompt

Control prompt	Line	Corpus prompt	Control prompt
Prompt control, orig	16	"PCL5": [int, int...] // 4 items, 0-4 scale	"PCL5": [int, int...] // 20 items, 0-4 scale

Table S1, continued

Control prompt	Line	Corpus prompt	Control prompt
Prompt control, no_pop	7	2. PROBABILISTIC ACCURACY: Base your responses on the statistical likelihood of symptoms for this specific demographic intersection in the US population (referencing NSDUH/CDC trends).	2. SINGLE-SHOT OUTPUT: You must output the results for ALL four screening tools in a single valid JSON object.
	8	3. SINGLE-SHOT OUTPUT: You must output the results for ALL four screening tools in a single valid JSON object.	3. NUMERIC ONLY: Do not generate text explanations. Return integers.
	9	4. NUMERIC ONLY: Do not generate text explanations. Return integers.	(absent)
	16	"PCL5": [int, int...] // 4 items, 0-4 scale	"PCL5": [int, int...] // 20 items, 0-4 scale
Prompt control, first_person	1	ROLE:	You are the person described in the profile below.
	2	You are a Clinical Simulation Engine designed for a synthetic data study.	Answer the four screening questionnaires as that person, about yourself.
	3	Your task is to generate PROBABILISTIC responses for a "Standardized Patient" based on a provided demographic profile.	(absent)
	5	OPERATIONAL CONSTRAINTS (STRICT):	CONSTRAINTS (STRICT):
	6	1. NO MORALIZING: Do not offer help, advice, or disclaimers. You are simulating data, not treating a patient.	1. NO MORALIZING: Do not offer help, advice, or disclaimers.
	7	2. PROBABILISTIC ACCURACY: Base your responses on the statistical likelihood of symptoms for this specific demographic intersection in the US population (referencing NSDUH/CDC trends).	2. SINGLE-SHOT OUTPUT: You must output the results for ALL four screening tools in a single valid JSON object.
	8	3. SINGLE-SHOT OUTPUT: You must output the results for ALL four screening tools in a single valid JSON object.	3. NUMERIC ONLY: Do not generate text explanations. Return integers.
	9	4. NUMERIC ONLY: Do not generate text explanations. Return integers.	(absent)
	16	"PCL5": [int, int...] // 4 items, 0-4 scale	"PCL5": [int, int...] // 20 items, 0-4 scale
Decoding control (both arms)	16	"PCL5": [int, int...] // 4 items, 0-4 scale	"PCL5": [int, int...] // 20 items, 0-4 scale

Note. Line numbers count lines of the corpus system prompt (S1.1). Lines not listed are identical. Computed with Python `difflib` on the prompt strings.

S2. Receipts

Script `scripts/86_receipts.py` checks every number in the article against its receipt file at the precision the article prints. Numbers that are derived by hand from receipt values are listed, with their derivation, in the script’s `DERIVED` register.

Table S2

Receipts: Every Number in the Article and the File That Prints It

Location	Claim	Receipt file	Script
sections/01_ introduction.tex:1	Prompted as a single Black woman on Medicaid with a household income under \$35,000, GPT-4o-mini returns a PHQ-8 total of 8 on its first clinical draw.	85_worked_case.csv	85_worked_case.py
sections/01_ introduction.tex:1	The same prompt run thirty times averages 10.3, above the usual screening cut point.	85_worked_case.csv	85_worked_case.py
sections/01_ introduction.tex:2	Adults in the national survey who share her race, sex, income band and marital status average 4.7.	85_worked_case.csv	85_worked_case.py
sections/01_ introduction.tex:3	We demonstrate the ladder on 28,800 PHQ-8 assessments from four models, against the National Health and Nutrition Examination Survey (NHANES).	corpus_v3_ provenance.csv	76_build_ corpus_v3.py; 87_supplement_ tables.py
sections/02_ ladder.tex:27	The level-2 rule in its present form was adopted on 25 September 2026, after an earlier rule had been run on the external datasets.	analysis_brm/L2.md	analysis note
sections/02_ ladder.tex:29	The level-4 invariance rule was written after the NHANES fits and before any model was fitted, with a margin of .05, and the margin was moved to .08 after the model results were in, because the reference itself does not hold on three of its six steps at .05.	analysis_brm/L4.md; l4_summary.csv	analysis note; 83a_controls_ receipt.py; 87_ supplement_ tables.py
sections/02_ ladder.tex:31	No verdict in this paper changes under either margin or under either status of the within-persona factor.	l4_summary.csv	83a_controls_ receipt.py; 87_ supplement_ tables.py
sections/02_ ladder.tex:62	Draws of one persona carry no order, and a crossed fit of persona by draw index finds no draw main effect, so the absolute and relative coefficients coincide.	gate_iteration_ check.csv	82a_gate_ gstudy.py
sections/02_ ladder.tex:63	Components are estimated from expected mean squares and checked against restricted maximum likelihood.	gate_reml_check.csv	82b_gate_reml_ check.py
sections/02_ ladder.tex:71	One point is about the size of the population’s sex gap on the PHQ-8, and half a point keeps a persona mean’s error below both that gap and half the low-to-high income gap.	gate_tolerance_ rationale.csv	82f_gate_ license.py

Location	Claim	Receipt file	Script
sections/02_ ladder.tex:104	On NHANES the largest subgroup departure is 1.36, so $\tau = 1.5$.	l1_personfit_ tolerance.csv	79c_l1_ personfit.py
sections/02_ ladder.tex:110	The row is descriptive because, with the corpus's totals kept, uniformly random answer vectors read as too prototypical on it (observed to expected 0.62), while person fit reads them as excess misfit (ratio 6.05).	l1_planted_ controls.csv	79e_l1_unit_ controls.py
sections/03_ data.tex:30	The clinical run took place between 28 December 2025 and 11 January 2026, and the narrative run on 9 and 10 January 2026.	corpus_v3_ provenance.csv	76_build_ corpus_v3.py; 87_supplement_ tables.py
sections/03_ data.tex:31	The narrative collection was a partial pass on 9 January that the same script completed on 10 January, so 1,104 GLM-4.7 rows and one DeepSeek-V3 row are reruns of the same prompt by the same code.	87_row_sources.csv	87_supplement_ tables.py
sections/03_ data.tex:33	The December clinical run left 1,532 calls without a valid answer, and recovery scripts filled them in January, 1,219 of them for GLM-4.7 and 313 for DeepSeek-V3.	corpus_v3_ provenance.csv	76_build_ corpus_v3.py; 87_supplement_ tables.py
sections/03_ data.tex:34	Of these, 125 GLM-4.7 rows used the corpus system prompt and 62 used a shortened system prompt without the population-reference instruction.	87_row_sources.csv	87_supplement_ tables.py
sections/03_ data.tex:35	No script on disk produced the other 1,345, so their prompt cannot be established, and Supplement S1 prints the recovery prompts that survive.	87_row_sources.csv	87_supplement_ tables.py
sections/03_ data.tex:36	A further 43 GPT-4o-mini rows were restored from a backup of the December run, so 1,575 clinical rows are not the December run's own output.	corpus_v3_ provenance.csv	76_build_ corpus_v3.py; 87_supplement_ tables.py
sections/03_ data.tex:37	Every rung was rerun on the December rows alone, and no per-model verdict changes at any rung (Supplement S3).	analysis_ brm/GATE.md; analysis_brm/L1.md; l2_verdicts.csv; 80d_ december_ summary.csv; l4_ summary_dec.csv	analysis note; analysis note; 78c_l2_r3_ verdicts.py; 78f_ l2_report_ tables.py; 83a_ controls_ receipt.py; 83g_ l2_conditional_ verdicts.py; 84_ fig_level2_ brm.py; 87_ supplement_ tables.py; 80d_ l3_summary.py; 87_supplement_ tables.py

Location	Claim	Receipt file	Script
sections/03_ data.tex:39	The corpus has 28,800 rows, of which 28,799 hold a valid PHQ-8.	corpus_v3_ provenance.csv; l1_ personfit_main.csv	76_build_ corpus_v3.py; 87_supplement_ tables.py; 79c_ l1_personfit.py; 79f_l1_ summary.py; 83a_controls_ receipt.py; 87_ supplement_ tables.py
sections/03_ data.tex:59	Note. Each model: 120 personas times 2 framings (clinical, narrative) times 30 draws. Decoding: provider default, nothing set, JSON output requested. Clinical run 28 December 2025 to 11 January 2026; narrative run 9–10 January 2026. One DeepSeek-V3 clinical draw returned an invalid PHQ-8 and is dropped. Persona attributes: race (5) times gender identity (4) times income (3) times relationship status (2).	corpus_v3_ provenance.csv	76_build_ corpus_v3.py; 87_supplement_ tables.py
sections/03_ data.tex:65	The two control runs used 12 single personas under the clinical framing, with 30 draws per cell.	../decoding_control_ design.json; ../prompt_control_ design.json	decoding_ control_io.py; 78d_l2_prompt_ control.py
sections/03_ data.tex:72	The decoding control’s design and primary outcomes were written to a dated file 21 minutes before its first call, and that file is in the release.	../decoding_control_ preregistration.json; ../decoding_control_ raw.jsonl	UNRESOLVED; UNRESOLVED
sections/03_ data.tex:76	The reference is NHANES 2005–2018 (cycles D to J): 36,274 adults aged 18 or over with all eight PHQ-8 items answered and a positive examination weight.	l1_nhanes_sample_ flow.csv; analysis_ brm/L4.md	79a_l1_grm_ fit.py; analysis note
sections/03_ data.tex:77	Estimates use the examination weights divided by seven and the masked design variables for strata and primary sampling units (105 strata, 214 PSUs, 109 design degrees of freedom).	l2_reference.csv; 80c_stdout.txt; analysis_brm/L2.md	78a_l2_ population_ reference.py; 78c_l2_r3_ verdicts.py; 78d_ l2_prompt_ control.py; 83a_ controls_ receipt.py; 83g_ l2_conditional_ verdicts.py; 37_ fetch_library.py; analysis note

Location	Claim	Receipt file	Script
sections/04_results.tex:14	The dependability of one draw runs from .21 for GPT-4o-mini under the narrative framing to .90 for Gemini-3-Flash.	gate_dstudy.csv	82a_gate_gstudy.py; 82d_gate_decoding_link.py; 82e_gate_december.py; 82f_gate_license.py; 87_supplement_tables.py
sections/04_results.tex:15	Gemini-3-Flash reaches .90 but its draws of one persona still spread with a standard deviation of about 1.5 points, above the 1-point limit.	gate_license_models.csv; gate_dstudy.csv	82f_gate_license.py; 87_supplement_tables.py; 82a_gate_gstudy.py; 82d_gate_decoding_link.py; 82e_gate_december.py; 82f_gate_license.py; 87_supplement_tables.py
sections/04_results.tex:17	Under the recommended rule, the number of draws needed is 9 for DeepSeek-V3, 10 for Gemini-3-Flash, 21 for GLM-4.7 and 35 for GPT-4o-mini.	gate_license_models.csv	82f_gate_license.py; 87_supplement_tables.py
sections/04_results.tex:18	GPT-4o-mini's narrative persona means reach a dependability of .887 at 30 draws, short of .90.	gate_dstudy.csv	82a_gate_gstudy.py; 82d_gate_decoding_link.py; 82e_gate_december.py; 82f_gate_license.py; 87_supplement_tables.py
sections/04_results.tex:19	Levels 2 and 3 read the persona mean over both framings, 60 draws, and for that mean GPT-4o-mini reaches .967.	gate_license_level3.csv	82f_gate_license.py
sections/04_results.tex:22	The draw-to-draw standard deviation is between 1.25 and 2.24 points in every model and framing.	gate_dstudy.csv	82a_gate_gstudy.py; 82d_gate_decoding_link.py; 82e_gate_december.py; 82f_gate_license.py; 87_supplement_tables.py

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sections/04_results.tex:23	The variance between personas ranges from 0.81 for GPT-4o-mini (narrative) to 20.69 for Gemini-3-Flash (narrative), a 26-fold range.	gate_dstudy.csv	82a_gate_gstudy.py; 82d_gate_decoding_link.py; 82e_gate_december.py; 82f_gate_license.py; 87_supplement_tables.py
sections/04_results.tex:24	Setting the temperature to 0 in the decoding control cut GPT-4o-mini's draw variance to 12.5% of its default value and did not raise the variance between personas.	gate_decoding_link.csv	82d_gate_decoding_link.py
sections/04_results.tex:25	At temperature 0 a single GPT-4o-mini draw reaches a dependability of only .75.	gate_decoding_link.csv	82d_gate_decoding_link.py
sections/04_results.tex:26	If a persona's score has to hold across the two prompt framings, DeepSeek-V3 and GPT-4o-mini cannot reach .90 with any number of draws (ceilings .83 and .88).	gate_dstudy_across_framing.csv	82a_gate_gstudy.py
sections/04_results.tex:28	Two clinical draws of one persona fall in different PHQ-8 severity categories 35.2% of the time [33.6, 36.8].	gate_bandflip_summary.csv	82c_gate_bandflip.py
sections/04_results.tex:30	A normal distribution with each cell's own mean and standard deviation, cut at the category boundaries, reproduces the cell flip rates with $R^2 = .88$, so flipping reflects how close a persona sits to a cut point as much as how much its draws vary.	gate_bandflip_regression.csv	82c_gate_bandflip.py
sections/04_results.tex:33	Her 30 clinical draws run from 7 to 16 with a standard deviation of 2.45, and two of them fall in different severity categories 60.5% of the time.	85_worked_case.csv	85_worked_case.py
sections/04_results.tex:34	Her first draw, 8, sits in the mild band, and her persona mean, 10.3, sits in the moderate band.	85_worked_case.csv	85_worked_case.py
sections/04_results.tex:43	NHANES puts 5% of adults below its within-total 5th percentile and 5% above its 95th, and the models put 0.0% to 1.0% below and 0.0% to 4.3% above.	l1_summary.csv	79f_l1_summary.py; 87_supplement_tables.py
sections/04_results.tex:46	No verdict differs between the two framings, and every model fails at every tolerance from 1.10 to 3.00.	l1_personfit_verdicts_tau.csv	79c_l1_personfit.py
sections/04_results.tex:47	At the loosest tolerance, 3.00, Gemini-3-Flash keeps only its misfit deficit and clinical GPT-4o-mini only its overfit deficit.	l1_personfit_verdicts_tau.csv	79c_l1_personfit.py
sections/04_results.tex:50	At matched totals, the models score 0.16 to 0.51 points higher than people on depressed mood.	l1_personfit_item_profile.csv	79c_l1_personfit.py
sections/04_results.tex:51	They score 0.22 to 1.11 points lower on fatigue, one of the somatic items people endorse first.	l1_personfit_item_profile.csv	79c_l1_personfit.py

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sections/04_results.tex:53	Two draws from different personas at the same total are identical vectors 4.6 to 39 times as often as two NHANES adults at that total.	l1_pattern_reuse.csv	79e_l1_unit_controls.py; 79f_l1_summary.py
sections/04_results.tex:54	On the gateway row, the models violate the PHQ-8's core-symptom rule at 0.5% to 21% of the human rate at the same totals.	l1_summary.csv	79f_l1_summary.py; 87_supplement_tables.py
sections/04_results.tex:57	None falls below the within-total 5th percentile, none falls above the 95th, and her mean probability integral transform is .40.	85_worked_case.csv	85_worked_case.py
sections/04_results.tex:58	None of her 32 elevated draws breaks the gateway rule.	85_worked_case.csv	85_worked_case.py
sections/04_results.tex:61	The verdicts hold on the December rows alone and under an unweighted calibration model.	l1_summary.csv	79f_l1_summary.py; 87_supplement_tables.py
sections/04_results.tex:62	Against the smaller NHANES 2021–2023 sample, DeepSeek-V3 and GPT-4o-mini move from compressed to an overfit deficit, because their misfit tails become too imprecise to place (Supplement S3).	l1_personfit_main.csv	79c_l1_personfit.py; 79f_l1_summary.py; 83a_controls_receipt.py; 87_supplement_tables.py
sections/04_results.tex:74	The gap between low and high income is steepened in every model, at 1.91 (DeepSeek-V3) to 5.07 (Gemini-3-Flash) times the population gap.	l2_headline.csv	78c_l2_r3_verdicts.py; 84_fig_level2_brm.py
sections/04_results.tex:75	Across the three income contrasts, 10 of the 12 per-model readings are steepened.	l2_headline.csv	78c_l2_r3_verdicts.py; 84_fig_level2_brm.py
sections/04_results.tex:76	The other two, Gemini-3-Flash's middle-to-high gap (1.31) and DeepSeek-V3's low-to-middle gap (1.33), read kept or steepened.	l2_headline.csv	78c_l2_r3_verdicts.py; 84_fig_level2_brm.py
sections/04_results.tex:77	On the marginal reference the income gaps are smaller relative to the population's, and Low minus High is still steepened for Gemini-3-Flash and GLM-4.7 and kept or steepened for the other two (Supplement S4).	l2_headline.csv	78c_l2_r3_verdicts.py; 84_fig_level2_brm.py
sections/04_results.tex:80	GPT-4o-mini and DeepSeek-V3 keep a quarter to two fifths of the population's sex gap (ratios 0.23 and 0.41), read missing or attenuated.	l2_headline.csv	78c_l2_r3_verdicts.py; 84_fig_level2_brm.py
sections/04_results.tex:81	GLM-4.7 doubles it (2.06), and Gemini-3-Flash is undetermined.	l2_headline.csv	78c_l2_r3_verdicts.py; 84_fig_level2_brm.py
sections/04_results.tex:82	The pooled ratio of 0.97 averages these opposite errors.	l2_headline.csv	78c_l2_r3_verdicts.py; 84_fig_level2_brm.py

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sections/04_results.tex:85	Among adults of equal income, sex and marital status, the NHANES Black–White gap is -0.28 points, while the crude gap is +0.33.	l2_reference.csv	78a_l2_population_reference.py; 78c_l2_r3_verdicts.py; 78d_l2_prompt_control.py; 83a_controls_receipt.py; 83g_l2_conditional_verdicts.py
sections/04_results.tex:86	The standardised Hispanic–White gap is -0.18, and its interval covers zero, so that contrast records no population gap for every model.	l2_stop.csv	78c_l2_r3_verdicts.py
sections/04_results.tex:87	For Black–White, GPT-4o-mini reads reversed or missing and the other three models are undetermined.	l2_headline.csv	78c_l2_r3_verdicts.py; 84_fig_level2_brm.py
sections/04_results.tex:88	For Asian–White, DeepSeek-V3 reads attenuated or kept and the other three are undetermined.	l2_headline.csv	78c_l2_r3_verdicts.py; 84_fig_level2_brm.py
sections/04_results.tex:93	With the draw-only standard error, 9 of the 35 verdicts become sharper and none moves outside its primary verdict.	83g_l2_conditional_verdicts.csv	83g_l2_conditional_verdicts.py
sections/04_results.tex:94	Under four other mappings of relationship status in the reference, 18 of the 112 per-model verdicts change, 16 of them on race or sex, and the low-to-high income gap stays steepened in every model (Supplement S3).	l2_verdicts.csv	78c_l2_r3_verdicts.py; 78f_l2_report_tables.py; 83a_controls_receipt.py; 83g_l2_conditional_verdicts.py; 84_fig_level2_brm.py; 87_supplement_tables.py
sections/04_results.tex:95	In the prompt control, the pooled low-to-high and low-to-middle income gaps stay steepened under all three system prompts, so the steepening does not come from the role or the population-reference instruction.	l2_prompt_control_verdicts.csv	78d_l2_prompt_control.py
sections/04_results.tex:96	With four persona pairs per model in that run, the per-model readings range from undetermined to steepened, and only Gemini-3-Flash’s low-to-high gap is steepened under every prompt.	l2_prompt_control_verdicts.csv	78d_l2_prompt_control.py
sections/04_results.tex:99	In the worked-case stratum (Black women who are single), GPT-4o-mini puts the low-income persona 3.4 points above the high-income one.	85_worked_case.csv	85_worked_case.py
sections/04_results.tex:100	In NHANES the same stratum differs by 1.6 points.	85_worked_case.csv	85_worked_case.py

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sections/04_ results.tex:108	After post-stratification the overall residual is +2.1 points for GLM-4.7, +2.6 for Gemini-3-Flash, +4.6 for DeepSeek-V3 and +4.7 for GPT-4o-mini.	80d_headline.csv	80d_l3_ summary.py
sections/04_ results.tex:109	All 40 model-by-group residuals are positive, with 90% intervals above zero.	80d_pass_counts.csv; 80d_ranges.csv	80d_l3_ summary.py; 80d_l3_ summary.py
sections/04_ results.tex:110	No model passes all ten groups at 0.2 SD, 1 point or 2 points.	80d_model_ verdicts.csv	80d_l3_ summary.py
sections/04_ results.tex:111	Of the 40 rows, none passes at 0.2 SD, one passes at 1 point, and six pass at 2 points.	80d_pass_counts.csv	80d_l3_ summary.py
sections/04_ results.tex:112	In every model the largest residual is in the low-income band.	80c_l3_results.csv	80c_l3_ estimates.py
sections/04_ results.tex:114	Against NHANES 2021–2023, with cohabiting adults counted as married in both cycles, every residual falls by 0.9 to 2.1 points and no model passes all ten groups at 2 points.	80d_era_shift_ permodel.csv; 80d_ model_verdicts_ all.csv	80d_l3_ summary.py; 80d_l3_ summary.py
sections/04_ results.tex:115	The overall rows of Gemini-3-Flash and GLM-4.7 pass at 2 points against that cycle, so a use that needs only the national level would pass these two models against 2021–2023 and fail them against 2005–2018.	80d_model_ verdicts_all.csv	80d_l3_ summary.py
sections/04_ results.tex:118	Pooled over models, 13.5% of simulated respondents score 10 or more against 7.0% in NHANES.	80d_headline.csv	80d_l3_ summary.py
sections/04_ results.tex:119	In the low-income band the simulated share is 55.5% against 13.6%.	80d_headline.csv	80d_l3_ summary.py
sections/04_ results.tex:120	In the middle and high income bands the simulated share is below the population's (4.4% against 7.3%, and 1.5% against 3.7%), although the means there are too high.	80d_headline.csv	80d_l3_ summary.py
sections/04_ results.tex:121	In 8 of the 50 primary rows (ten groups for each model and for the pooled population) the mean is significantly too high and the prevalence significantly too low.	80d_sign_ discordance_ primary.csv	80d_l3_ summary.py
sections/04_ results.tex:122	DeepSeek-V3's overall prevalence passes the ratio band (1.05) only because a low-income excess (32.7% against 13.6%) offsets shortfalls in the middle and high bands.	80d_headline.csv; 80d_model_ verdicts.csv	80d_l3_ summary.py; 80d_l3_ summary.py
sections/04_ results.tex:123	The worked-case persona's mean of 10.0 over both framings sits 5.3 points above her own NHANES cell (4.7).	85_worked_case.csv	85_worked_ case.py
sections/04_ results.tex:124	Against adults on Medicaid with household income under \$35,000, the label her prompt carried, the mean of all such adults is 5.5.	85_worked_case.csv	85_worked_ case.py
sections/04_ results.tex:133	Their ratios of the first to the second eigenvalue are 1.09 to 1.73, against 7.47 in NHANES, and each has negative loadings.	14_structure.csv	83a_controls_ receipt.py
sections/04_ results.tex:134	The rule fails in every one of 1,000 persona bootstrap replicates for both models.	14_structure.csv	83a_controls_ receipt.py

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sections/04_results.tex:138	Their congruence with the NHANES loadings is .990 to .998.	l4_structure.csv	83a_controls_receipt.py
sections/04_results.tex:139	An equal-loading vector scores .998 against the same loadings, so R2 adds little beyond R1 on this scale.	l4_nhanes_selfphi.csv	81b_l4_structure.py
sections/04_results.tex:141	In NHANES, income is invariant at the .08 margin at both steps.	l4_invariance.csv	81c_l4_invariance.py; 87_supplement_tables.py
sections/04_results.tex:142	In the two models the metric step's RMSEA_D is .16 to .20, with every lower 90% limit above .13.	l4_invariance.csv	81c_l4_invariance.py; 87_supplement_tables.py
sections/04_results.tex:143	At the .08 margin, seven of the eight sex steps in these two models are indeterminate, because 24 to 30 personas per group cannot resolve them, and seven of the eight race steps hold, as they do in NHANES.	l4_invariance.csv; l4_invariance_pattern.csv	81c_l4_invariance.py; 87_supplement_tables.py; 81c_l4_invariance.py; 81e_l4_summary.py
sections/04_results.tex:144	The verdicts are the same at the .05 margin.	l4_summary.csv	83a_controls_receipt.py; 87_supplement_tables.py
sections/04_results.tex:147	In NHANES, 3.5% of item variance lies between demographic cells, and the covariance within cells carries the general factor on its own ($\lambda_1/\lambda_2 = 4.33$).	l4_between_within.csv	81b_l4_structure.py; 81e_l4_summary.py
sections/04_results.tex:148	In Gemini-3-Flash and GLM-4.7, 48% to 64% of item variance lies between personas, and among draws of one persona there is no general factor (λ_1/λ_2 of 1.29 to 1.59).	l4_between_within.csv	81b_l4_structure.py; 81e_l4_summary.py
sections/04_results.tex:153	That value comes from pooling two models that have a general factor with two that do not.	l4_group_congruence.csv	81b_l4_structure.py
sections/04_results.tex:154	Within each model that has a factor, the income congruence is .961 to .990, and the failure rests on the invariance test instead.	l4_group_congruence.csv	81b_l4_structure.py
sections/05_controls.tex:16	Four independent pseudo-models form a panel, and there are 200 replicates.	analysis_brm/CONTROLS.md	analysis note
sections/05_controls.tex:19	Run on the real corpus with the full reference, the control code reproduces every published rung number, 303 of 303 checks.	83a_receipt.csv	83a_controls_receipt.py
sections/05_controls.tex:22	The pseudo-model built from NHANES respondents passes level 1 in 99.4% of 800 model replicates.	83c_l1.csv	83c_controls_summary.py
sections/05_controls.tex:23	It passes level 3 on all ten groups in 100% of replicates at 2 points and 98.1% at 1 point.	83c_l3.csv	83c_controls_summary.py
sections/05_controls.tex:24	It passes the level-4 structure rules (R1, R2 and the R3 diagnostic, in both framings) in all 200 replicates.	83c_l4.csv	83c_controls_summary.py

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sections/05_controls.tex:26	At level 2, a determinate wrong verdict occurs in at most 1.0% of readings at any dose and scope.	83c_l2.csv	83c_controls_summary.py
sections/05_controls.tex:27	The level-2 intervals cover the true ratio in .986 to 1.000 of replicates per model, against .986 nominal.	83c_l2_coverage.csv	83c_controls_summary.py
sections/05_controls.tex:32	At level 2 the control's reference is a half sample, with twice the variance of the full survey, and per-model detection needs a threefold gap.	83c_l2.csv	83c_controls_summary.py
sections/05_controls.tex:33	Two models' real income ratios (1.9 and 2.2) lie below that dose, and against the full reference the rung still names both steepened.	l2_headline.csv	78c_l2_r3_verdicts.py; 84_fig_level2_brm.py
sections/05_controls.tex:54	Note. 200 replicates of a four-model panel of NHANES pseudo-models. Doses are nested, so a draw altered at one dose is altered at every larger dose. Per-dose detection rates for every rung and tolerance are in Supplement S5.	83c_l1.csv; 83c_l2.csv; 83c_l3.csv; 83c_l4.csv; analysis_brm/L1.md	83c_controls_summary.py; 83c_controls_summary.py; 83c_controls_summary.py; 83c_controls_summary.py; analysis note
sections/05_controls.tex:60	In a simulation at the corpus's own precision, a faithful model is named kept alone in at most 37% of readings for the low-to-high income gap and 20% for the sex gap, and in under 1% for every other contrast.	83f_l2_parametric_power.csv	83f_l2_parametric_power.py
sections/05_controls.tex:61	These are the most favourable settings, the draw-only standard error at the most precise model, and under the primary pairs standard error neither rate exceeds 2%.	83f_l2_parametric_power.csv	83f_l2_parametric_power.py
sections/05_controls.tex:63	A twofold steepening of an income or sex gap is named correctly in 87% to 100% of draws under the draw-only standard error.	83f_l2_parametric_power.csv	83f_l2_parametric_power.py
sections/05_controls.tex:64	At the audit's own precision, the racial contrasts of a faithful model are stopped in 46% (Black–White) and 74% (Hispanic–White) of per-model readings.	83c_l2.csv	83c_controls_summary.py
sections/05_controls.tex:69	With every draw decorrelated, Tucker's congruence still averages .98, and R2 holds in 20% to 25% of framings while R1 and R3 fail in all of them.	83c_l4.csv	83c_controls_summary.py
sections/05_controls.tex:71	Under the persona design the positive control's misfit ratio averages 1.15, because the design gives equal weight to cells and over-represents subgroups with more misfit.	83c_l1.csv	83c_controls_summary.py
sections/05_controls.tex:72	A tolerance of 1.25 would therefore fail real people drawn through the same design, and the misfit tail passes at tau = 1.5 in 99.4% of replicates.	83c_l1.csv	83c_controls_summary.py

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sections/06_external.tex:15	The 154 group contrasts that the earlier version of this protocol had read as 51 kept and 85 missing become, under the ladder's rule, 89 undetermined, 29 attenuated or kept, 17 missing or attenuated, 12 kept or steepened, 5 reversed or missing, 1 kept and 1 steepened.	l2_external_r3_summary.csv; l2_external_r3.csv; analysis_brm/L2.md	78e_l2_external_r3.py; 78e_l2_external_r3.py; analysis note
sections/06_external.tex:16	Most of the earlier "missing" readings become undetermined (48) or two-region verdicts (37), because their intervals span more than one region.	l2_external_r3_receipts.csv	78e_l2_external_r3.py
sections/06_external.tex:21	In the main run, 23 of 61 contrasts record no population gap and 12 record a reference too imprecise to read.	l2_external_r3.csv; analysis_brm/L2.md	78e_l2_external_r3.py; analysis note
sections/06_external.tex:22	The two Republican–Democrat contrasts on ideology and on the Trump two-party vote share are kept (both 0.92).	l2_external_r3.csv	78e_l2_external_r3.py
sections/06_external.tex:23	The Black–White gap in Trump vote share is attenuated (0.53), as is the Republican–Democrat gap in Trump's share among all voters (0.60).	l2_external_r3.csv	78e_l2_external_r3.py
sections/06_external.tex:24	Black–White and Republican–Democrat differences in patriotism are missing.	l2_external_r3.csv	78e_l2_external_r3.py
sections/08_limitations.tex:10	Of the clinical rows, 1,575 are not the December run's own output, 1,345 of them came from a prompt that cannot be established, and no per-model verdict changes without them.	corpus_v3_provenance.csv; 87_row_sources.csv	76_build_corpus_v3.py; 87_supplement_tables.py; 87_supplement_tables.py
sections/08_limitations.tex:20	In the worked example, four models generated 28,800 PHQ-8 assessments and every model failed every rung above the gate, while a positive control built from survey respondents passed and each rung detected planted failures.	corpus_v3_provenance.csv	76_build_corpus_v3.py; 87_supplement_tables.py
figures/fig2_level2_note.tex:4	The pooled Women–Men interval (0.746 to 1.262) crosses both edges of the kept band, so it is undetermined.	l2_verdicts.csv	78c_l2_r3_verdicts.py; 78f_l2_report_tables.py; 83a_controls_receipt.py; 83g_l2_conditional_verdicts.py; 84_fig_level2_brm.py; 87_supplement_tables.py
figures/fig2_level2_note.tex:7	Each model has 12 persona pairs per racial contrast, 24 for sex and 16 per income contrast.}	l2_headline.csv	78c_l2_r3_verdicts.py; 84_fig_level2_brm.py

Location	Claim	Receipt file	Script
tables/tab_ gate.tex:16	DeepSeek-V3 & Clinical & 2.14 & 2.03 & .51 & .97 & 4 & 9 & 27.0	gate_dstudy.csv; gate_ components.csv; gate_bandflip_ summary.csv; gate_ license_models.csv	82a_gate_ gstudy.py; 82d_ gate_decoding_ link.py; 82e_ gate_ december.py; 82f_gate_ license.py; 87_ supplement_ tables.py; 82a_ gate_gstudy.py; 82b_gate_reml_ check.py; 82c_ gate_bandflip.py; 82f_gate_ license.py; 87_ supplement_ tables.py
tables/tab_ gate.tex:17	& Narrative & 1.61 & 1.57 & .51 & .97 & 4 & 9 & 22.4	gate_dstudy.csv; gate_ components.csv; gate_bandflip_ summary.csv; gate_ license_models.csv	82a_gate_ gstudy.py; 82d_ gate_decoding_ link.py; 82e_ gate_ december.py; 82f_gate_ license.py; 87_ supplement_ tables.py; 82a_ gate_gstudy.py; 82b_gate_reml_ check.py; 82c_ gate_bandflip.py; 82f_gate_ license.py; 87_ supplement_ tables.py

Location	Claim	Receipt file	Script
tables/tab_ gate.tex:18	Gemini-3-Flash & Clinical & 18.74 & 2.17 & .90 & >.99 & 3 & 9 & 31.6	gate_dstudy.csv; gate_ components.csv; gate_bandflip_ summary.csv; gate_ license_models.csv	82a_gate_ gstudy.py; 82d_ gate_decoding_ link.py; 82e_ gate_ december.py; 82f_gate_ license.py; 87_ supplement_ tables.py; 82a_ gate_gstudy.py; 82b_gate_reml_ check.py; 82c_ gate_bandflip.py; 82f_gate_ license.py; 87_ supplement_ tables.py
tables/tab_ gate.tex:19	& Narrative & 20.69 & 2.27 & .90 & >.99 & 3 & 10 & 28.7	gate_dstudy.csv; gate_ components.csv; gate_bandflip_ summary.csv; gate_ license_models.csv	82a_gate_ gstudy.py; 82d_ gate_decoding_ link.py; 82e_ gate_ december.py; 82f_gate_ license.py; 87_ supplement_ tables.py; 82a_ gate_gstudy.py; 82b_gate_reml_ check.py; 82c_ gate_bandflip.py; 82f_gate_ license.py; 87_ supplement_ tables.py

Location	Claim	Receipt file	Script
tables/tab_ gate.tex:20	GPT-4o-mini & Clinical & 1.89 & 3.06 & .38 & .95 & 7 & 15 & 37.3	gate_dstudy.csv; gate_ components.csv; gate_bandflip_ summary.csv; gate_ license_models.csv	82a_gate_ gstudy.py; 82d_ gate_decoding_ link.py; 82e_ gate_ december.py; 82f_gate_ license.py; 87_ supplement_ tables.py; 82a_ gate_gstudy.py; 82b_gate_reml_ check.py; 82c_ gate_bandflip.py; 82f_gate_ license.py; 87_ supplement_ tables.py
tables/tab_ gate.tex:21	& Narrative & 0.81 & 3.09 & .21 & .89 & 16 & 35 & 39.9	gate_dstudy.csv; gate_ components.csv; gate_bandflip_ summary.csv; gate_ license_models.csv	82a_gate_ gstudy.py; 82d_ gate_decoding_ link.py; 82e_ gate_ december.py; 82f_gate_ license.py; 87_ supplement_ tables.py; 82a_ gate_gstudy.py; 82b_gate_reml_ check.py; 82c_ gate_bandflip.py; 82f_gate_ license.py; 87_ supplement_ tables.py

Location	Claim	Receipt file	Script
tables/tab_ gate.tex:22	GLM-4.7 & Clinical & 15.04 & 5.04 & .75 & .99 & 6 & 21 & 44.8	gate_dstudy.csv; gate_ components.csv; gate_bandflip_ summary.csv; gate_ license_models.csv	82a_gate_ gstudy.py; 82d_ gate_decoding_ link.py; 82e_ gate_ december.py; 82f_gate_ license.py; 87_ supplement_ tables.py; 82a_ gate_gstudy.py; 82b_gate_reml_ check.py; 82c_ gate_bandflip.py; 82f_gate_ license.py; 87_ supplement_ tables.py
tables/tab_ gate.tex:23	& Narrative & 20.16 & 3.55 & .85 & .99 & 4 & 15 & 37.6	gate_dstudy.csv; gate_ components.csv; gate_bandflip_ summary.csv; gate_ license_models.csv	82a_gate_ gstudy.py; 82d_ gate_decoding_ link.py; 82e_ gate_ december.py; 82f_gate_ license.py; 87_ supplement_ tables.py; 82a_ gate_gstudy.py; 82b_gate_reml_ check.py; 82c_ gate_bandflip.py; 82f_gate_ license.py; 87_ supplement_ tables.py

Location	Claim	Receipt file	Script
tables/tab_ l1.tex:16	DeepSeek-V3 & Clinical & 0.3 & 0.01–0.21 & 0.0 & 0.00–0.01 & .36 & Compressed	l1_summary.csv; l1_personfit_main.csv; l1_gateway_main.csv; l1_pattern_reuse.csv	79f_l1_summary.py; 87_supplement_tables.py; 79c_l1_personfit.py; 79f_l1_summary.py; 83a_controls_receipt.py; 87_supplement_tables.py; 79b_l1_gateway.py; 79f_l1_summary.py; 79e_l1_unit_controls.py; 79f_l1_summary.py
tables/tab_ l1.tex:17	& Narrative & 0.1 & 0.00–0.11 & 0.0 & 0.00–0.01 & .37 & Compressed	l1_summary.csv; l1_personfit_main.csv; l1_gateway_main.csv; l1_pattern_reuse.csv	79f_l1_summary.py; 87_supplement_tables.py; 79c_l1_personfit.py; 79f_l1_summary.py; 83a_controls_receipt.py; 87_supplement_tables.py; 79b_l1_gateway.py; 79f_l1_summary.py; 79e_l1_unit_controls.py; 79f_l1_summary.py
tables/tab_ l1.tex:18	Gemini-3-Flash & Clinical & 0.0 & 0.00–0.00 & 0.8 & 0.11–0.39 & .66 & Compressed	l1_summary.csv; l1_personfit_main.csv; l1_gateway_main.csv; l1_pattern_reuse.csv	79f_l1_summary.py; 87_supplement_tables.py; 79c_l1_personfit.py; 79f_l1_summary.py; 83a_controls_receipt.py; 87_supplement_tables.py; 79b_l1_gateway.py; 79f_l1_summary.py; 79e_l1_unit_controls.py; 79f_l1_summary.py

Location	Claim	Receipt file	Script
tables/tab_ l1.tex:19	& Narrative & 0.0 & 0.00–0.00 & 0.9 & 0.11–0.42 & .65 & Compressed	l1_summary.csv; l1_ personfit_main.csv; l1_gateway_ main.csv; l1_ pattern_reuse.csv	79f_l1_ summary.py; 87_ supplement_ tables.py; 79c_ l1_personfit.py; 79f_l1_ summary.py; 83a_controls_ receipt.py; 87_ supplement_ tables.py; 79b_ l1_gateway.py; 79f_l1_ summary.py; 79e_l1_unit_ controls.py; 79f_ l1_summary.py
tables/tab_ l1.tex:20	GPT-4o-mini & Clinical & 1.0 & 0.09–0.37 & 0.0 & 0.00–0.03 & .38 & Compressed	l1_summary.csv; l1_ personfit_main.csv; l1_gateway_ main.csv; l1_ pattern_reuse.csv	79f_l1_ summary.py; 87_ supplement_ tables.py; 79c_ l1_personfit.py; 79f_l1_ summary.py; 83a_controls_ receipt.py; 87_ supplement_ tables.py; 79b_ l1_gateway.py; 79f_l1_ summary.py; 79e_l1_unit_ controls.py; 79f_ l1_summary.py
tables/tab_ l1.tex:21	& Narrative & 0.8 & 0.09–0.30 & 0.0 & 0.00–0.03 & .38 & Compressed	l1_summary.csv; l1_ personfit_main.csv; l1_gateway_ main.csv; l1_ pattern_reuse.csv	79f_l1_ summary.py; 87_ supplement_ tables.py; 79c_ l1_personfit.py; 79f_l1_ summary.py; 83a_controls_ receipt.py; 87_ supplement_ tables.py; 79b_ l1_gateway.py; 79f_l1_ summary.py; 79e_l1_unit_ controls.py; 79f_ l1_summary.py

Location	Claim	Receipt file	Script
tables/tab_ l1.tex:22	GLM-4.7 & Clinical & 0.1 & 0.01–0.04 & 3.4 & 0.55–0.92 & .65 & Misfit deficit	l1_summary.csv; l1_ personfit_main.csv; l1_gateway_ main.csv; l1_ pattern_reuse.csv	79f_l1_ summary.py; 87_ supplement_ tables.py; 79c_ l1_personfit.py; 79f_l1_ summary.py; 83a_controls_ receipt.py; 87_ supplement_ tables.py; 79b_ l1_gateway.py; 79f_l1_ summary.py; 79e_l1_unit_ controls.py; 79f_ l1_summary.py
tables/tab_ l1.tex:23	& Narrative & 0.1 & 0.00–0.04 & 4.3 & 0.74–1.15 & .65 & Misfit deficit	l1_summary.csv; l1_ personfit_main.csv; l1_gateway_ main.csv; l1_ pattern_reuse.csv	79f_l1_ summary.py; 87_ supplement_ tables.py; 79c_ l1_personfit.py; 79f_l1_ summary.py; 83a_controls_ receipt.py; 87_ supplement_ tables.py; 79b_ l1_gateway.py; 79f_l1_ summary.py; 79e_l1_unit_ controls.py; 79f_ l1_summary.py
tables/tab_ l1.tex:25	NHANES & & 5.0 & & 5.0 & & .50 &	l1_summary.csv; l1_ personfit_main.csv; l1_gateway_ main.csv; l1_ pattern_reuse.csv	79f_l1_ summary.py; 87_ supplement_ tables.py; 79c_ l1_personfit.py; 79f_l1_ summary.py; 83a_controls_ receipt.py; 87_ supplement_ tables.py; 79b_ l1_gateway.py; 79f_l1_ summary.py; 79e_l1_unit_ controls.py; 79f_ l1_summary.py

Location	Claim	Receipt file	Script
tables/tab_ l3.tex:16	DeepSeek-V3 & +4.58 [4.49, 4.67] & +3.47 & +4.06 to +5.13 & 0 / 0 / 0 & 7.35 & 7.01	80d_headline.csv; 80d_headline_by_ model.csv; 80d_ pass_counts.csv; 80c_l3_results.csv	80d_l3_ summary.py; 80d_l3_ summary.py; 80d_l3_ summary.py; 80c_l3_ estimates.py
tables/tab_ l3.tex:17	Gemini-3-Flash & +2.59 [2.49, 2.69] & +1.36 & +1.33 to +7.37 & 0 / 0 / 2 & 16.42 & 7.01	80d_headline.csv; 80d_headline_by_ model.csv; 80d_ pass_counts.csv; 80c_l3_results.csv	80d_l3_ summary.py; 80d_l3_ summary.py; 80d_l3_ summary.py; 80c_l3_ estimates.py
tables/tab_ l3.tex:18	GPT-4o-mini & +4.68 [4.57, 4.79] & +3.57 & +4.23 to +5.32 & 0 / 0 / 0 & 16.92 & 7.01	80d_headline.csv; 80d_headline_by_ model.csv; 80d_ pass_counts.csv; 80c_l3_results.csv	80d_l3_ summary.py; 80d_l3_ summary.py; 80d_l3_ summary.py; 80c_l3_ estimates.py
tables/tab_ l3.tex:19	GLM-4.7 & +2.12 [2.00, 2.24] & +0.91 & +0.66 to +6.40 & 0 / 1 / 4 & 13.23 & 7.01	80d_headline.csv; 80d_headline_by_ model.csv; 80d_ pass_counts.csv; 80c_l3_results.csv	80d_l3_ summary.py; 80d_l3_ summary.py; 80d_l3_ summary.py; 80c_l3_ estimates.py
tables/tab_ l3.tex:21	Pooled (descriptive) & +3.49 [3.42, 3.57] & +2.33 & +2.59 to +6.05 & 0 / 0 / 0 & 13.48 & 7.01	80d_headline.csv; 80d_headline_by_ model.csv; 80d_ pass_counts.csv; 80c_l3_results.csv	80d_l3_ summary.py; 80d_l3_ summary.py; 80d_l3_ summary.py; 80c_l3_ estimates.py

Location	Claim	Receipt file	Script
tables/tab_ l4.tex:17	DeepSeek-V3 & Clinical & 1.73 & -.72 & Fail & .349 [.285] & (Fail) & – & Fail & .17 & 1.45	l4_summary.csv; l4_ structure.csv; l4_ between_within.csv; l4_invariance_ pattern.csv; l4_ nhanes_selfphi.csv	83a_controls_ receipt.py; 87_ supplement_ tables.py; 83a_ controls_ receipt.py; 81b_ l4_structure.py; 81e_l4_ summary.py; 81c_l4_ invariance.py; 81e_l4_ summary.py; 81b_l4_ structure.py
tables/tab_ l4.tex:18	& Narrative & 1.43 & -.68 & Fail & .163 [.074] & (Fail) & – & Fail & .14 & 1.34	l4_summary.csv; l4_ structure.csv; l4_ between_within.csv; l4_invariance_ pattern.csv; l4_ nhanes_selfphi.csv	83a_controls_ receipt.py; 87_ supplement_ tables.py; 83a_ controls_ receipt.py; 81b_ l4_structure.py; 81e_l4_ summary.py; 81c_l4_ invariance.py; 81e_l4_ summary.py; 81b_l4_ structure.py
tables/tab_ l4.tex:19	Gemini-3-Flash & Clinical & 8.69 & .71 & Pass & .990 [.986] & Pass & Fail & Fail & .63 & 1.29	l4_summary.csv; l4_ structure.csv; l4_ between_within.csv; l4_invariance_ pattern.csv; l4_ nhanes_selfphi.csv	83a_controls_ receipt.py; 87_ supplement_ tables.py; 83a_ controls_ receipt.py; 81b_ l4_structure.py; 81e_l4_ summary.py; 81c_l4_ invariance.py; 81e_l4_ summary.py; 81b_l4_ structure.py

Location	Claim	Receipt file	Script
tables/tab_ l4.tex:20	& Narrative & 9.33 & .69 & Pass & .990 [.987] & Pass & Fail & Fail & .64 & 1.45	l4_summary.csv; l4_ structure.csv; l4_ between_within.csv; l4_invariance_ pattern.csv; l4_ nhanes_selfphi.csv	83a_controls_ receipt.py; 87_ supplement_ tables.py; 83a_ controls_ receipt.py; 81b_ l4_structure.py; 81e_l4_ summary.py; 81c_l4_ invariance.py; 81e_l4_ summary.py; 81b_l4_ structure.py
tables/tab_ l4.tex:21	GPT-4o-mini & Clinical & 1.09 & -.24 & Fail & .364 [.178] & (Fail) & – & Fail & .10 & 1.23	l4_summary.csv; l4_ structure.csv; l4_ between_within.csv; l4_invariance_ pattern.csv; l4_ nhanes_selfphi.csv	83a_controls_ receipt.py; 87_ supplement_ tables.py; 83a_ controls_ receipt.py; 81b_ l4_structure.py; 81e_l4_ summary.py; 81c_l4_ invariance.py; 81e_l4_ summary.py; 81b_l4_ structure.py
tables/tab_ l4.tex:22	& Narrative & 1.33 & -.57 & Fail & .043 [-.017] & (Fail) & – & Fail & .06 & 1.26	l4_summary.csv; l4_ structure.csv; l4_ between_within.csv; l4_invariance_ pattern.csv; l4_ nhanes_selfphi.csv	83a_controls_ receipt.py; 87_ supplement_ tables.py; 83a_ controls_ receipt.py; 81b_ l4_structure.py; 81e_l4_ summary.py; 81c_l4_ invariance.py; 81e_l4_ summary.py; 81b_l4_ structure.py

Location	Claim	Receipt file	Script
tables/tab_ l4.tex:23	GLM-4.7 & Clinical & 5.73 & .66 & Pass & .996 [.994] & Pass & Fail & Fail & .48 & 1.59	l4_summary.csv; l4_ structure.csv; l4_ between_within.csv; l4_invariance_ pattern.csv; l4_ nhanes_selfphi.csv	83a_controls_ receipt.py; 87_ supplement_ tables.py; 83a_ controls_ receipt.py; 81b_ l4_structure.py; 81e_l4_ summary.py; 81c_l4_ invariance.py; 81e_l4_ summary.py; 81b_l4_ structure.py
tables/tab_ l4.tex:24	& Narrative & 8.12 & .75 & Pass & .998 [.997] & Pass & Fail & Fail & .59 & 1.34	l4_summary.csv; l4_ structure.csv; l4_ between_within.csv; l4_invariance_ pattern.csv; l4_ nhanes_selfphi.csv	83a_controls_ receipt.py; 87_ supplement_ tables.py; 83a_ controls_ receipt.py; 81b_ l4_structure.py; 81e_l4_ summary.py; 81c_l4_ invariance.py; 81e_l4_ summary.py; 81b_l4_ structure.py
tables/tab_ l4.tex:26	NHANES & & 7.47 & .67 & & & & & .035 & 4.33	l4_summary.csv; l4_ structure.csv; l4_ between_within.csv; l4_invariance_ pattern.csv; l4_ nhanes_selfphi.csv	83a_controls_ receipt.py; 87_ supplement_ tables.py; 83a_ controls_ receipt.py; 81b_ l4_structure.py; 81e_l4_ summary.py; 81c_l4_ invariance.py; 81e_l4_ summary.py; 81b_l4_ structure.py

S3. Corpus Provenance and Sensitivity Analyses

S3.1 Row Sources

Table S3 counts the corpus rows by framing, model and source. Of the clinical rows, 1,345 are in-place recovery fills (313 DeepSeek-V3 and 1,032 GLM-4.7), 125 come from `slow_recovery.py`, 62 come from `last_mile_recovery_opt.py`, and 43 GPT-4o-mini rows were restored from a backup of the December run. These 1,575 clinical rows are not the December run’s own output. Of the narrative rows, 1,104 GLM-4.7 rows and 1 DeepSeek-V3 row were produced on 10 January by the same script and prompt that produced the first pass on 9 January. The corpus has 28,800 rows, of which 28,799 hold a valid PHQ-8.

Table S3

Corpus Rows by Framing, Model and Source

Framing	Model	row_source	narr_source	Description	Rows	Valid PHQ-8
Clinical	DeepSeek-V3	dec28_main	–	December run, own output	3,287	3,286
	DeepSeek-V3	recovery_inplace	–	Recovery, filled in place	313	313
	Gemini-3-Flash	dec28_main	–	December run, own output	3,600	3,600
	GPT-4o-mini	dec28_main	–	December run, own output	3,557	3,557
	GPT-4o-mini	dec28_restored	–	December run, restored from backup	43	43
	GLM-4.7	dec28_main	–	December run, own output	2,381	2,381
	GLM-4.7	recovery_inplace	–	Recovery, filled in place	1,032	1,032
	GLM-4.7	recovery_last_mile	–	Recovery, <code>last_mile_recovery_opt.py</code>	62	62
	GLM-4.7	recovery_slow	–	Recovery, <code>slow_recovery.py</code>	125	125
Narrative	DeepSeek-V3	narrative	narr_main	Narrative run, first pass	3,599	3,599
	DeepSeek-V3	narrative	narr_rerun_same_script	Narrative run, rerun, same script	1	1
	Gemini-3-Flash	narrative	narr_main	Narrative run, first pass	3,600	3,600
	GPT-4o-mini	narrative	narr_main	Narrative run, first pass	3,600	3,600
	GLM-4.7	narrative	narr_main	Narrative run, first pass	2,496	2,496
	GLM-4.7	narrative	narr_rerun_same_script	Narrative run, rerun, same script	1,104	1,104
Total					28,800	28,799

Note. Counts from `data/model_outputs_v3.csv`, grouped by script 87 and checked against `corpus_v3_provenance.csv` (script 76).

S3.2 December-Only Corpus

The December-only corpus keeps the clinical rows labelled `dec28_main` and `dec28_restored` and every narrative row. Each rung was rerun on it. Script 87 counts the

per-model verdicts that differ from the full corpus: 0 at level 1, 0 at level 2, 0 at level 3 and 0 at level 4. At level 2, 1 pooled verdict differs (Table S6). At the gate, 3 values of the minimum or recommended k change, all for GLM-4.7 clinical, and 0 of them cross 30, the number of draws per persona in the corpus (Table S4). At level 3, the largest change in a group residual across framings is 0.75 points, and 0.17 points among rows that lose no design cell; 40 rows lose at least one cell (80d_december_summary.csv).

Table S4

Gate on the Full Corpus and on December Rows Only

Outcome	Model	Framing	Personas		$\phi(1)$		Minimum k		Recommended k	
			Full	Dec.	Full	Dec.	Full	Dec.	Full	Dec.
PHQ-8 total	DeepSeek-V3	Clinical	120	111	.514	.513	4	4	9	9
	DeepSeek-V3	Narrative	120	120	.505	.505	4	4	9	9
	Gemini-3-Flash	Clinical	120	120	.896	.896	3	3	9	9
	Gemini-3-Flash	Narrative	120	120	.901	.901	3	3	10	10
	GPT-4o-mini	Clinical	120	120	.382	.382	7	7	15	15
	GPT-4o-mini	Narrative	120	120	.207	.207	16	16	35	35
	GLM-4.7	Clinical	120	80	.749	.715	6	6	21	23
	GLM-4.7	Narrative	120	120	.850	.850	4	4	15	15
PHQ-8 ≥ 10	DeepSeek-V3	Clinical	120	111	.427	.424	12	12	45	45
	DeepSeek-V3	Narrative	120	120	.294	.294	10	10	34	34
	Gemini-3-Flash	Clinical	120	120	.745	.745	7	7	26	26
	Gemini-3-Flash	Narrative	120	120	.792	.792	6	6	21	21
	GPT-4o-mini	Clinical	120	120	.252	.252	16	16	63	63
	GPT-4o-mini	Narrative	120	120	.119	.119	30	30	71	71
	GLM-4.7	Clinical	120	80	.584	.550	10	11	39	41
	GLM-4.7	Narrative	120	120	.698	.698	8	8	29	29

Note. Dec. = December rows only (dec28_main, dec28_restored and all narrative rows). Minimum: the larger of the k for $\phi \geq .80$ and for $SE \leq 1.0$. Recommended: the larger of the k for $\phi \geq .90$ and for $SE \leq 0.5$.

Table S5

Level 1 on the Full Corpus and on December Rows Only

Model	Framing	Misfit ratio		Overfit ratio		Verdict, full	Verdict, Dec.
		Full	Dec.	Full	Dec.		
DeepSeek-V3	Both	0.05	0.05	0.00	0.00	compressed	compressed
DeepSeek-V3	Clinical	0.07	0.07	0.00	0.00	compressed	compressed
DeepSeek-V3	Narrative	0.03	0.03	0.00	0.00	compressed	compressed
Gemini-3-Flash	Both	0.00	0.00	0.17	0.17	compressed	compressed
Gemini-3-Flash	Clinical	0.00	0.00	0.16	0.16	compressed	compressed

Table S5, continued

Model	Framing	Misfit ratio		Overfit ratio		Verdict, full	Verdict, Dec.
		Full	Dec.	Full	Dec.		
Gemini-3-Flash	Narrative	0.00	0.00	0.18	0.18	compressed	compressed
GPT-4o-mini	Both	0.18	0.18	0.00	0.00	compressed	compressed
GPT-4o-mini	Clinical	0.20	0.20	0.00	0.00	compressed	compressed
GPT-4o-mini	Narrative	0.17	0.17	0.00	0.00	compressed	compressed
GLM-4.7	Both	0.02	0.01	0.77	0.78	misfit deficit	misfit deficit
GLM-4.7	Clinical	0.02	0.02	0.68	0.64	misfit deficit	misfit deficit
GLM-4.7	Narrative	0.01	0.01	0.87	0.87	misfit deficit	misfit deficit
Pooled	Both	0.06	0.06	0.24	0.22	compressed	compressed
Pooled	Clinical	0.07	0.08	0.21	0.16	compressed	compressed
Pooled	Narrative	0.05	0.05	0.26	0.26	compressed	compressed

Note. Reference: NHANES 2005–2018, weighted graded response model; all personas. Every verdict is a fail; the verdict column gives the failure reading, where compressed means a deficit in both tails.

Table S6

Level 2 on the Full Corpus and on December Rows Only

Estimand	Contrast	Model	Full corpus		December only		Changed
			Ratio	Verdict	Ratio	Verdict	
Standardised	Black minus White	DeepSeek-V3	−0.75	undetermined	−0.82	undetermined	
		Gemini-3-Flash	1.39	undetermined	1.39	undetermined	
		GPT-4o-mini	−0.76	reversed or missing	−0.76	reversed or missing	
		GLM-4.7	1.96	undetermined	1.96	undetermined	
		Pooled	0.46	reference too imprecise	0.44	reference too imprecise	
	Hispanic minus White	DeepSeek-V3	−0.95	no population gap	−0.95	no population gap	
		Gemini-3-Flash	0.04	no population gap	0.04	no population gap	
		GPT-4o-mini	−0.77	no population gap	−0.77	no population gap	
		GLM-4.7	2.42	no population gap	3.33	no population gap	
		Pooled	0.19	no population gap	0.41	no population gap	
	Asian minus White	DeepSeek-V3	0.74	attenuated or kept	0.74	attenuated or kept	
		Gemini-3-Flash	1.52	undetermined	1.52	undetermined	
		GPT-4o-mini	0.42	undetermined	0.42	undetermined	
		GLM-4.7	1.82	undetermined	1.82	undetermined	
		Pooled	1.13	undetermined	1.13	undetermined	
	Women minus Men	DeepSeek-V3	0.41	missing or attenuated	0.37	missing or attenuated	
		Gemini-3-Flash	1.17	undetermined	1.17	undetermined	
		GPT-4o-mini	0.23	missing or attenuated	0.23	missing or attenuated	
		GLM-4.7	2.06	steepened	2.00	steepened	
		Pooled	0.97	undetermined	0.94	attenuated or kept	*
	Low minus High SES	DeepSeek-V3	1.91	steepened	1.94	steepened	
		Gemini-3-Flash	5.07	steepened	5.07	steepened	
		GPT-4o-mini	2.16	steepened	2.16	steepened	
		GLM-4.7	5.03	steepened	5.14	steepened	
		Pooled	3.54	steepened	3.58	steepened	
	Middle minus High SES	DeepSeek-V3	2.57	steepened	2.65	steepened	

Table S6, continued

Estimand	Contrast	Model	Full corpus		December only		Changed
			Ratio	Verdict	Ratio	Verdict	
Marginal	Low minus Middle SES	Gemini-3-Flash	1.31	kept or steepened	1.31	kept or steepened	
		GPT-4o-mini	2.25	steepened	2.25	steepened	
		GLM-4.7	2.21	steepened	2.25	steepened	
		Pooled	2.08	steepened	2.11	steepened	
		DeepSeek-V3	1.33	kept or steepened	1.32	kept or steepened	
		Gemini-3-Flash	8.38	steepened	8.38	steepened	
		GPT-4o-mini	2.08	steepened	2.08	steepened	
		GLM-4.7	7.51	steepened	7.68	steepened	
		Pooled	4.83	steepened	4.86	steepened	
	Black minus White	DeepSeek-V3	0.65	reference too imprecise	0.71	reference too imprecise	
		Gemini-3-Flash	-1.20	undetermined	-1.20	undetermined	
		GPT-4o-mini	0.66	reference too imprecise	0.66	reference too imprecise	
		GLM-4.7	-1.69	reversed	-1.69	reversed	
		Pooled	-0.40	reversed or missing	-0.38	reversed or missing	
	Hispanic minus White	DeepSeek-V3	0.72	reference too imprecise	0.72	reference too imprecise	
		Gemini-3-Flash	-0.03	undetermined	-0.03	undetermined	
		GPT-4o-mini	0.58	reference too imprecise	0.58	reference too imprecise	
		GLM-4.7	-1.83	reversed or missing	-2.52	reversed or missing	
		Pooled	-0.14	undetermined	-0.31	undetermined	
	Asian minus White	DeepSeek-V3	0.94	reference too imprecise	0.94	reference too imprecise	
		Gemini-3-Flash	1.92	undetermined	1.92	undetermined	
		GPT-4o-mini	0.53	undetermined	0.53	undetermined	
		GLM-4.7	2.30	undetermined	2.30	undetermined	
		Pooled	1.42	undetermined	1.42	undetermined	
	Women minus Men	DeepSeek-V3	0.37	missing or attenuated	0.34	missing or attenuated	
		Gemini-3-Flash	1.05	undetermined	1.05	undetermined	
		GPT-4o-mini	0.20	missing or attenuated	0.20	missing or attenuated	
		GLM-4.7	1.84	steepened	1.79	steepened	
		Pooled	0.87	attenuated or kept	0.85	attenuated or kept	
	Low minus High SES	DeepSeek-V3	1.23	kept or steepened	1.25	kept or steepened	
		Gemini-3-Flash	3.26	steepened	3.26	steepened	
		GPT-4o-mini	1.39	kept or steepened	1.39	kept or steepened	
		GLM-4.7	3.23	steepened	3.30	steepened	
		Pooled	2.28	steepened	2.30	steepened	
	Middle minus High SES	DeepSeek-V3	1.75	steepened	1.81	steepened	
		Gemini-3-Flash	0.89	attenuated or kept	0.89	attenuated or kept	
		GPT-4o-mini	1.53	kept or steepened	1.53	kept or steepened	
		GLM-4.7	1.50	kept or steepened	1.53	kept or steepened	
		Pooled	1.42	kept or steepened	1.44	kept or steepened	
	Low minus Middle SES	DeepSeek-V3	0.82	attenuated or kept	0.80	attenuated or kept	
		Gemini-3-Flash	5.13	steepened	5.13	steepened	
		GPT-4o-mini	1.27	kept or steepened	1.27	kept or steepened	
		GLM-4.7	4.60	steepened	4.70	steepened	
		Pooled	2.95	steepened	2.98	steepened	

Note. Rows of 12_verdicts.csv with analysis headline and dec_only. Ratio = simulated gap over population gap. * = verdict differs.

Table S7*Level 3 on the Full Corpus and on December Rows Only*

Model	Overall residual		Group residuals, range		Largest change	All groups pass at 2 points	
	Full	Dec.	Full	Dec.		Full	Dec.
DeepSeek-V3	4.58	4.55	4.06 to 5.13	4.06 to 5.10	0.25	No	No
Gemini-3-Flash	2.59	2.59	1.33 to 7.37	1.33 to 7.37	0.00	No	No
GPT-4o-mini	4.68	4.68	4.23 to 5.32	4.23 to 5.32	0.00	No	No
GLM-4.7	2.12	2.15	0.66 to 6.40	0.67 to 6.41	0.75	No	No
Pooled	3.49	3.50	2.59 to 6.05	2.61 to 6.07	0.15	No	No

Note. Specification S1, post-stratified estimand, framings combined, PHQ-8 mean in points. Largest change = largest absolute change in a group residual across the ten groups.

Table S8*Level 4 on the Full Corpus and on December Rows Only*

Model	Framing	λ_1/λ_2		Tucker ϕ		R4 at .08		Level 4 pass	
		Full	Dec.	Full	Dec.	Full	Dec.	Full	Dec.
DeepSeek-V3	Clinical	1.73	1.72	.349	.351	not interpretable	not interpretable	No	No
DeepSeek-V3	Narrative	1.43	1.43	.163	.163	not interpretable	not interpretable	No	No
Gemini-3-Flash	Clinical	8.69	8.69	.990	.990	fail	fail	No	No
Gemini-3-Flash	Narrative	9.33	9.33	.990	.990	fail	fail	No	No
GPT-4o-mini	Clinical	1.09	1.09	.364	.364	not interpretable	not interpretable	No	No
GPT-4o-mini	Narrative	1.33	1.33	.043	.043	not interpretable	not interpretable	No	No
GLM-4.7	Clinical	5.73	5.23	.996	.994	fail	fail	No	No
GLM-4.7	Narrative	8.12	8.12	.998	.998	fail	fail	No	No

Note. R1 needs $\lambda_1/\lambda_2 \geq 3$ and every loading $\geq .30$; R2 reads the lower 90% limit of Tucker's ϕ against .95.

S3.3 NHANES 2021–2023 Reference

The gate reads no population reference, so it has no 2021–2023 version. Level 1 was refitted with a graded response model on NHANES 2021–2023 (Table S9). Under that reference the level-1 verdicts that change, on both framings pooled, are: DeepSeek-V3, compressed to overfit deficit; GPT-4o-mini, compressed to overfit deficit. Level 3 was rerun against NHANES 2021–2023 and the 2017–2020 pre-pandemic file, each with its own design (Tables S10 and S11). Against 2021–2023 the per-model overall residuals move by -2.09 to -0.93 points (mean -1.16), and the per-model group residuals range from -0.33 to 6.27 points. Level 2 and level 4 were not run against NHANES 2021–2023.

Table S9*Level 1 Against NHANES 2005–2018 and NHANES 2021–2023*

Model	Framing	Misfit ratio		Overfit ratio		Verdict, 2005–18	Verdict, 2021–23
		2005–18	2021–23	2005–18	2021–23		
DeepSeek-V3	Both	0.05	0.28	0.00	0.00	compressed	overfit deficit
DeepSeek-V3	Clinical	0.07	0.34	0.00	0.00	compressed	overfit deficit
DeepSeek-V3	Narrative	0.03	0.22	0.00	0.00	compressed	overfit deficit
Gemini-3-Flash	Both	0.00	0.26	0.17	0.12	compressed	compressed
Gemini-3-Flash	Clinical	0.00	0.33	0.16	0.11	compressed	compressed
Gemini-3-Flash	Narrative	0.00	0.18	0.18	0.13	compressed	compressed
GPT-4o-mini	Both	0.18	0.99	0.00	0.00	compressed	overfit deficit
GPT-4o-mini	Clinical	0.20	1.00	0.00	0.00	compressed	overfit deficit
GPT-4o-mini	Narrative	0.17	0.97	0.00	0.00	compressed	overfit deficit
GLM-4.7	Both	0.02	0.06	0.77	0.83	misfit deficit	misfit deficit
GLM-4.7	Clinical	0.02	0.07	0.68	0.74	misfit deficit	misfit deficit
GLM-4.7	Narrative	0.01	0.05	0.87	0.91	misfit deficit	misfit deficit
pooled	Both	0.06	0.40	0.24	0.24	compressed	overfit deficit
pooled	Clinical	0.07	0.44	0.21	0.21	compressed	overfit deficit
pooled	Narrative	0.05	0.36	0.26	0.26	compressed	overfit deficit

Note. Each reference has its own weighted graded response model. Every verdict is a fail; the verdict column gives the failure reading, where compressed means a deficit in both tails.

Table S10*Level 3 Overall Residual Under Three NHANES Reference Periods*

Model	2005–2018 (S1)			2017–2020 (S6)			2021–2023 (S7)		
	Overall	Row	All	Overall	Row	All	Overall	Row	All
DeepSeek-V3	4.58	No	No	4.30	No	No	3.47	No	No
Gemini-3-Flash	2.59	No	No	2.22	No	No	1.36	Yes	No
GPT-4o-mini	4.68	No	No	4.41	No	No	3.57	No	No
GLM-4.7	2.12	No	No	1.74	Yes	No	0.91	Yes	No
Pooled	3.49	No	No	3.17	No	No	2.33	No	No

Note. Overall = post-stratified overall residual (simulated minus reference, PHQ-8 points), framings combined. Row = the overall row passes at 2 points. All = all ten group rows pass at 2 points.

Table S11*Level 3 Pooled Residual by Group Under Three Reference Periods*

Group	Reference mean			Pooled residual		
	2005–18	2017–20	2021–23	2005–18	2021–23	Shift
Overall	2.89	3.10	3.96	3.49	2.33	–1.17
White	2.86	3.15	3.91	3.26	2.17	–1.09
Black	3.20	3.10	4.21	4.11	2.99	–1.12
Asian	2.16	2.08	3.10	3.34	2.16	–1.18
Hispanic	3.13	3.22	4.32	4.33	2.63	–1.70

Table S11, continued

Group	Reference mean			Pooled residual		
	2005–18	2017–20	2021–23	2005–18	2021–23	Shift
Men	2.43	2.59	3.44	3.43	2.31	−1.12
Women	3.34	3.57	4.47	3.55	2.35	−1.21
Low	4.14	4.39	5.24	6.05	4.95	−1.10
Middle	3.07	3.43	4.33	3.16	1.91	−1.24
High	2.18	2.35	3.15	2.59	1.62	−0.98

Note. PHQ-8 points, post-stratified, pooled over models and framings. Shift = residual under 2021–2023 minus residual under 2005–2018.

S3.4 Marital-Status Codings

The primary reference counts married adults as married and widowed, divorced, separated and never-married adults as single, leaves cohabiting adults out, and counts 18- and 19-year-olds as single. Level 3 was rerun with cohabiting adults counted as married (S2) and as single (S2b) (Table S12). Level 2 was rerun with four other references: every marital status kept (all marital), single read as never married, 18- and 19-year-olds dropped, and marital status left out of the standardisation (Table S13). Across those four references, 18 per-model standardised verdict codes differ from the primary.

Table S12

Level 3 Under Three Codings of Marital Status

Model	S1: cohabiting dropped			S2: as married			S2b: as single		
	Overall	Max $ R $	Pass	Overall	Max $ R $	Pass	Overall	Max $ R $	Pass
DeepSeek-V3	4.58	5.13	No	4.55	5.08	No	4.59	5.13	No
Gemini-3-Flash	2.59	7.37	No	2.59	7.23	No	2.66	7.39	No
GPT-4o-mini	4.68	5.32	No	4.65	5.25	No	4.68	5.30	No
GLM-4.7	2.12	6.40	No	2.10	6.24	No	2.20	6.41	No
Pooled	3.49	6.05	No	3.47	5.94	No	3.53	6.05	No

Note. Post-stratified residual in PHQ-8 points, framings combined. Max $|R|$ = largest absolute group residual over ten groups. Pass = all ten group rows pass at 2 points.

Table S13

Level 2 Standardised Ratios Under Alternative Marital-Status References

Contrast	Model	Primary	All marital	Never married	Drop 18–19	No marital
Black minus White	DeepSeek-V3	−0.75 u	−0.71 u	−1.14 N	−0.75 u	−1.01 u

Table S13, continued

Contrast	Model	Primary	All marital	Never married	Drop 18–19	No marital
	Gemini-3-Flash	1.39 u	1.31 u	2.10 N	1.38 u	1.86 u
	GPT-4o-mini	−0.76 r/m	−0.72 r/m	−1.15 N	−0.76 r/m	−1.02 u
	GLM-4.7	1.96 u	1.85 u	2.97 N	1.96 u	2.63 u
	Pooled	0.46 X	0.43 X	0.70 N	0.46 X	0.62 X
Hispanic minus White	DeepSeek-V3	−0.95 N	−0.62 u	−1.15 N	−0.91 N	−0.60 u
	Gemini-3-Flash	0.04 N	0.03 u	0.05 N	0.04 N	0.02 u
	GPT-4o-mini	−0.77 N	−0.50 r/m	−0.93 N	−0.73 N	−0.49 r/m
	GLM-4.7	2.42 N	1.59 u	2.93 N	2.31 N	1.54 u
Asian minus White	Pooled	0.19 N	0.12 u	0.22 N	0.18 N	0.12 u
	DeepSeek-V3	0.74 a/k	0.72 a/k	0.81 u	0.72 a/k	0.68 a/k
	Gemini-3-Flash	1.52 u	1.47 u	1.65 u	1.48 u	1.38 u
	GPT-4o-mini	0.42 u	0.41 u	0.46 u	0.41 u	0.38 m/a
Women minus Men	GLM-4.7	1.82 u	1.76 u	1.98 u	1.77 u	1.66 u
	Pooled	1.13 u	1.09 u	1.22 u	1.10 u	1.03 u
	DeepSeek-V3	0.41 m/a	0.40 m/a	0.45 m/a	0.41 m/a	0.39 m/a
	Gemini-3-Flash	1.17 u	1.13 u	1.28 k/s	1.17 u	1.11 u
Low minus High SES	GPT-4o-mini	0.23 m/a	0.22 m/a	0.25 m/a	0.23 m/a	0.22 m/a
	GLM-4.7	2.06 s	1.98 s	2.25 s	2.06 s	1.96 s
	Pooled	0.97 u	0.93 a/k	1.06 k/s	0.97 u	0.92 a/k
	DeepSeek-V3	1.91 s	1.87 s	2.12 s	1.83 s	1.66 s
Middle minus High SES	Gemini-3-Flash	5.07 s	4.96 s	5.62 s	4.85 s	4.42 s
	GPT-4o-mini	2.16 s	2.11 s	2.39 s	2.06 s	1.88 s
	GLM-4.7	5.03 s	4.92 s	5.58 s	4.81 s	4.39 s
	Pooled	3.54 s	3.47 s	3.93 s	3.39 s	3.09 s
Low minus Middle SES	DeepSeek-V3	2.57 s	2.55 s	3.13 s	2.50 s	2.30 s
	Gemini-3-Flash	1.31 k/s	1.30 k/s	1.60 k/s	1.27 k/s	1.17 k/s
	GPT-4o-mini	2.25 s	2.23 s	2.75 s	2.20 s	2.02 s
	GLM-4.7	2.21 s	2.19 s	2.70 s	2.15 s	1.98 k/s
	Pooled	2.08 s	2.07 s	2.54 s	2.03 s	1.87 s
	DeepSeek-V3	1.33 k/s	1.29 k/s	1.37 k/s	1.25 k/s	1.14 u
	Gemini-3-Flash	8.38 s	8.10 s	8.59 s	7.87 s	7.13 s
	GPT-4o-mini	2.08 s	2.01 s	2.13 s	1.95 s	1.77 s
	GLM-4.7	7.51 s	7.26 s	7.70 s	7.06 s	6.39 s
	Pooled	4.83 s	4.67 s	4.95 s	4.53 s	4.11 s

Note. Each cell: ratio and verdict code. Codes: **s** steepened, **k/s** kept or steepened, **a/k** attenuated or kept, **m/a** missing or attenuated, **r** reversed, **r/m** reversed or missing, **u** undetermined, **X** reference too imprecise, **N** no population gap. Rows of `12_verdicts.csv` by analysis.

S3.5 Income-Anchor Sensitivity

The persona labels state dollar incomes and insurance, and the primary reference bands the NHANES ratio of family income to the poverty line. The sensitivity mapping uses household income and insurance type on the 2007–2018 cycles, where NHANES top-codes household income. Table S14 gives the level-3 income-group residuals under each mapping, Table S15 the marginal residuals, and Table S16 the level-2 income contrasts.

Table S14

Level 3 Income-Group Residuals Under Four Income Mappings

Model	S1: poverty ratio		S3: PIR 2007–18		S4: persona string		S5: top-coded	
	Low, Mid, High	Pass	Low, Mid, High	Pass	Low, Mid, High	Pass	Low, Mid, High	Pass
DeepSeek-V3	5.13, 4.92, 4.06	No	5.08, 4.84, 4.02	No	3.98, 5.43, 4.21	No	3.98, 5.43, 4.27	No
Gemini-3-Flash	7.37, 1.33, 1.42	No	7.32, 1.25, 1.37	No	6.32, 1.71, 1.59	No	6.32, 1.71, 1.65	No
GPT-4o-mini	5.32, 4.88, 4.23	No	5.26, 4.79, 4.19	No	4.13, 5.32, 4.43	No	4.13, 5.32, 4.50	No
GLM-4.7	6.40, 1.50, 0.66	No	6.36, 1.43, 0.61	No	5.45, 1.77, 0.79	No	5.45, 1.77, 0.85	No
Pooled	6.05, 3.16, 2.59	No	6.01, 3.08, 2.55	No	4.97, 3.56, 2.75	No	4.97, 3.56, 2.82	No

Note. Post-stratified residuals in PHQ-8 points, framings combined. S3 uses the poverty ratio on 2007–2018 only; S4 maps the persona’s dollar income and insurance onto household income and insurance type; S5 is S4 with the high band top-coded. Pass = all ten group rows pass at 2 points.

Table S15

Marginal Income-Group Residuals Under Two Income Mappings

Model	Low		Middle		High	
	S3	S4	S3	S4	S3	S4
DeepSeek-V3	5.08	3.98	4.84	5.43	4.02	4.21
Gemini-3-Flash	7.32	6.32	1.25	1.71	1.37	1.59
GPT-4o-mini	5.26	4.13	4.79	5.32	4.19	4.43
GLM-4.7	6.36	5.45	1.43	1.77	0.61	0.79

Note. Marginal residual (simulated group mean minus reference group mean), PHQ-8 points, from 80d_income_anchor_permodel.csv.

Table S16

Level 2 Income Contrasts Under the Alternative Income Mapping

Estimand	Contrast	Model	Poverty ratio (primary)		Persona string	
			Ratio	Verdict	Ratio	Verdict
Standardised	Low minus High SES	DeepSeek-V3	1.91	steepened	1.04	kept or steepened
		Gemini-3-Flash	5.07	steepened	2.76	steepened
		GPT-4o-mini	2.16	steepened	1.18	kept or steepened

Table S16, continued

Estimand	Contrast	Model	Poverty ratio (primary)		Persona string	
			Ratio	Verdict	Ratio	Verdict
		GLM-4.7	5.03	steepened	2.74	steepened
		Pooled	3.54	steepened	1.93	steepened
		DeepSeek-V3	2.57	steepened	3.29	steepened
		Gemini-3-Flash	1.31	kept or steepened	1.68	kept or steepened
	Middle minus High SES	GPT-4o-mini	2.25	steepened	2.89	steepened
		GLM-4.7	2.21	steepened	2.83	steepened
		Pooled	2.08	steepened	2.67	steepened
		DeepSeek-V3	1.33	kept or steepened	0.48	attenuated
	Low minus Middle SES	Gemini-3-Flash	8.38	steepened	3.03	steepened
		GPT-4o-mini	2.08	steepened	0.75	attenuated or kept
		GLM-4.7	7.51	steepened	2.72	steepened
		Pooled	4.83	steepened	1.75	steepened
Marginal	Low minus High SES	DeepSeek-V3	1.23	kept or steepened	0.70	attenuated or kept
		Gemini-3-Flash	3.26	steepened	1.86	steepened
		GPT-4o-mini	1.39	kept or steepened	0.79	attenuated or kept
		GLM-4.7	3.23	steepened	1.85	steepened
		Pooled	2.28	steepened	1.30	kept or steepened
		DeepSeek-V3	1.75	steepened	3.07	steepened
		Gemini-3-Flash	0.89	attenuated or kept	1.56	kept or steepened
		GPT-4o-mini	1.53	kept or steepened	2.69	steepened
	Middle minus High SES	GLM-4.7	1.50	kept or steepened	2.64	steepened
		Pooled	1.42	kept or steepened	2.49	steepened
		DeepSeek-V3	0.82	attenuated or kept	0.30	missing or attenuated
		Gemini-3-Flash	5.13	steepened	1.91	steepened
	Low minus Middle SES	GPT-4o-mini	1.27	kept or steepened	0.47	attenuated
		GLM-4.7	4.60	steepened	1.71	steepened
		Pooled	2.95	steepened	1.10	kept or steepened

Note. Rows of 12_verdicts.csv with analysis headline and ref_income_alt. The alternative mapping was run for the marginal estimand on the income contrasts only.

S4. Full Per-Model Results

S4.1 Gate

Tables S17 and S18 give every gate quantity with its persona-bootstrap interval, for the PHQ-8 total and for the indicator of a total of 10 or more.

Table S17

Gate: Variance Components and Dependability With Persona-Bootstrap Intervals

Outcome	Model	Framing	σ_p^2	σ_r^2	$\phi(1)$	$\phi(30)$
PHQ-8 total	DeepSeek-V3	Clinical	2.14 [1.58, 2.71]	2.03 [1.78, 2.28]	.513 [.449, .563]	.969 [.961, .975]
	DeepSeek-V3	Narrative	1.61 [1.16, 2.09]	1.57 [1.39, 1.78]	.505 [.439, .561]	.968 [.959, .975]
	Gemini-3-Flash	Clinical	18.74 [15.78, 21.32]	2.17 [1.90, 2.42]	.896 [.870, .916]	.996 [.995, .997]
	Gemini-3-Flash	Narrative	20.69 [17.78, 23.42]	2.27 [2.02, 2.54]	.901 [.877, .919]	.996 [.995, .997]
	GPT-4o-mini	Clinical	1.89 [1.51, 2.23]	3.06 [2.85, 3.27]	.382 [.332, .419]	.949 [.937, .956]
	GPT-4o-mini	Narrative	0.81 [0.62, 0.98]	3.08 [2.88, 3.32]	.207 [.168, .243]	.887 [.858, .906]
	GLM-4.7	Clinical	15.04 [12.10, 17.66]	5.04 [4.56, 5.53]	.749 [.697, .787]	.989 [.986, .991]
PHQ-8 ≥ 10	GLM-4.7	Narrative	20.16 [16.68, 23.27]	3.55 [3.25, 3.88]	.850 [.814, .874]	.994 [.992, .995]
	DeepSeek-V3	Clinical	0.08 [0.06, 0.10]	0.11 [0.10, 0.13]	.426 [.349, .499]	.957 [.941, .968]
	DeepSeek-V3	Narrative	0.04 [0.02, 0.05]	0.08 [0.07, 0.10]	.294 [.224, .360]	.926 [.897, .944]
	Gemini-3-Flash	Clinical	0.19 [0.17, 0.20]	0.06 [0.05, 0.08]	.745 [.677, .802]	.989 [.984, .992]
	Gemini-3-Flash	Narrative	0.20 [0.18, 0.21]	0.05 [0.04, 0.07]	.792 [.730, .845]	.991 [.988, .994]
	GPT-4o-mini	Clinical	0.05 [0.04, 0.06]	0.16 [0.14, 0.17]	.252 [.210, .291]	.910 [.889, .925]
	GPT-4o-mini	Narrative	0.02 [0.02, 0.03]	0.18 [0.16, 0.19]	.119 [.092, .143]	.801 [.752, .834]
	GLM-4.7	Clinical	0.14 [0.12, 0.15]	0.10 [0.08, 0.12]	.584 [.507, .652]	.977 [.969, .983]
	GLM-4.7	Narrative	0.16 [0.14, 0.18]	0.07 [0.05, 0.09]	.698 [.620, .765]	.986 [.980, .990]

Note. Brackets: 95% persona-bootstrap intervals. σ_p^2 = variance between personas, σ_r^2 = variance between draws of one persona (squared PHQ-8 points for the total, squared proportion for the indicator). ϕ from the nested design.

Table S18

Gate: Draws Needed per Persona With Persona-Bootstrap Intervals

Outcome	Model	Framing	Dependability		Standard error		Rule	
			$\phi \geq .80$	$\phi \geq .90$	SE ≤ 1	SE ≤ 0.5	Min.	Rec.
PHQ-8 total	DeepSeek-V3	Clinical	4 [4, 5]	9 [7, 12]	3 [2, 3]	9 [8, 10]	4	9
	DeepSeek-V3	Narrative	4 [4, 6]	9 [8, 12]	2 [2, 2]	7 [6, 8]	4	9
	Gemini-3-Flash	Clinical	1 [1, 1]	2 [1, 2]	3 [2, 3]	9 [8, 10]	3	9
	Gemini-3-Flash	Narrative	1 [1, 1]	1 [1, 2]	3 [3, 3]	10 [9, 11]	3	10
	GPT-4o-mini	Clinical	7 [6, 9]	15 [13, 19]	4 [3, 4]	13 [12, 14]	7	15
	GPT-4o-mini	Narrative	16 [13, 20]	35 [29, 45]	4 [3, 4]	13 [12, 14]	16	35
	GLM-4.7	Clinical	2 [2, 2]	4 [3, 4]	6 [5, 6]	21 [19, 23]	6	21
	GLM-4.7	Narrative	1 [1, 1]	2 [2, 3]	4 [4, 4]	15 [13, 16]	4	15
PHQ-8 ≥ 10	DeepSeek-V3	Clinical	6 [5, 8]	13 [10, 17]	12 [10, 13]	45 [39, 52]	12	45
	DeepSeek-V3	Narrative	10 [8, 14]	22 [16, 32]	9 [7, 11]	34 [28, 41]	10	34
	Gemini-3-Flash	Clinical	2 [1, 2]	4 [3, 5]	7 [5, 9]	26 [20, 33]	7	26
	Gemini-3-Flash	Narrative	2 [1, 2]	3 [2, 4]	6 [4, 7]	21 [16, 28]	6	21
	GPT-4o-mini	Clinical	12 [10, 16]	27 [22, 34]	16 [14, 17]	63 [56, 68]	16	63
	GPT-4o-mini	Narrative	30 [24, 40]	67 [54, 90]	18 [17, 20]	71 [66, 77]	30	71
	GLM-4.7	Clinical	3 [3, 4]	7 [5, 9]	10 [9, 12]	39 [33, 47]	10	39
	GLM-4.7	Narrative	2 [2, 3]	4 [3, 6]	8 [6, 9]	29 [22, 36]	8	29

Note. Brackets: 95% persona-bootstrap intervals for k . Standard-error tolerances are in PHQ-8 points for the total and are applied on the same scale to the indicator as stored in `gate_dstudy.csv`. Min. = larger of the first and third columns; Rec. = larger of the second and fourth; both checked against `gate_license_models.csv`.

S4.2 Level 2

Table S19 gives the standardised estimand, which is primary, and Table S20 the marginal estimand. Each table carries the verdict under both stops and, as a sensitivity column, the verdict under the population-only stop.

Table S19*Level 2 per Model, Standardised Estimand*

Contrast	Model	Pop. gap	Sim. gap	Ratio	98.6% interval	Verdict	Population-only stop
Black minus White	DeepSeek-V3	−0.28	0.21	−0.75	[−3.74, 0.36]	undetermined	reference too imprecise
	Gemini-3-Flash		−0.39	1.39	[−0.61, 6.84]	undetermined	reference too imprecise
	GPT-4o-mini		0.22	−0.76	[−3.63, 0.19]	reversed or missing	reference too imprecise
	GLM-4.7		−0.56	1.96	[0.29, 8.75]	undetermined	reference too imprecise
	Pooled		−0.13	0.46	[−0.22, 2.34]	reference too imprecise	reference too imprecise
Hispanic minus White	DeepSeek-V3	−0.18	0.17	−0.95	unbounded	no population gap	no population gap
	Gemini-3-Flash		−0.01	0.04	unbounded	no population gap	no population gap
	GPT-4o-mini		0.14	−0.77	unbounded	no population gap	no population gap
	GLM-4.7		−0.44	2.42	unbounded	no population gap	no population gap
	Pooled		−0.03	0.19	unbounded	no population gap	no population gap
Asian minus White	DeepSeek-V3	−0.95	−0.71	0.74	[0.36, 1.20]	attenuated or kept	attenuated or kept
	Gemini-3-Flash		−1.45	1.52	[−0.39, 3.54]	undetermined	undetermined
	GPT-4o-mini		−0.40	0.42	[0.09, 0.79]	undetermined	undetermined
	GLM-4.7		−1.74	1.82	[0.55, 3.25]	undetermined	undetermined
	Pooled		−1.07	1.13	[0.57, 1.80]	undetermined	undetermined
Women minus Men	DeepSeek-V3	0.84	0.35	0.41	[0.23, 0.62]	missing or attenuated	missing or attenuated
	Gemini-3-Flash		0.99	1.17	[0.73, 1.70]	undetermined	undetermined
	GPT-4o-mini		0.19	0.23	[0.00, 0.47]	missing or attenuated	missing or attenuated
	GLM-4.7		1.73	2.06	[1.50, 2.76]	steepened	steepened
	Pooled		0.81	0.97	[0.75, 1.26]	undetermined	undetermined
Low minus High SES	DeepSeek-V3	1.30	2.48	1.91	[1.55, 2.39]	steepened	steepened
	Gemini-3-Flash		6.58	5.07	[3.85, 6.60]	steepened	steepened
	GPT-4o-mini		2.80	2.16	[1.78, 2.68]	steepened	steepened
	GLM-4.7		6.53	5.03	[4.05, 6.33]	steepened	steepened
	Pooled		4.60	3.54	[2.94, 4.38]	steepened	steepened
Middle minus High SES	DeepSeek-V3	0.61	1.56	2.57	[1.82, 3.94]	steepened	reference too imprecise
	Gemini-3-Flash		0.79	1.31	[0.86, 2.06]	kept or steepened	reference too imprecise
	GPT-4o-mini		1.37	2.25	[1.57, 3.49]	steepened	reference too imprecise
	GLM-4.7		1.34	2.21	[1.32, 3.62]	steepened	reference too imprecise
	Pooled		1.26	2.08	[1.52, 3.17]	steepened	reference too imprecise

Table S19, continued

Contrast	Model	Pop. gap	Sim. gap	Ratio	98.6% interval	Verdict	Population-only stop
Low minus Middle SES	DeepSeek-V3	0.69	0.92	1.33	[0.83, 2.09]	kept or steepened	reference too imprecise
	Gemini-3-Flash		5.79	8.38	[5.88, 12.59]	steepened	reference too imprecise
	GPT-4o-mini		1.44	2.08	[1.55, 3.04]	steepened	reference too imprecise
	GLM-4.7		5.19	7.51	[5.62, 10.96]	steepened	reference too imprecise
	Pooled		3.33	4.83	[3.62, 7.03]	steepened	reference too imprecise

Note. Gaps in PHQ-8 points. Ratio = simulated gap over population gap, with its Fieller interval at the Bonferroni level for 7 contrasts (one-sided $\alpha = .0071$). Verdict: the primary rule, whose reference-imprecision stop fires when the observed simulated gap, treated as measured without error, would still give a ratio interval spanning three or more regions. Population-only stop: the verdict when that stop instead fires whenever the population gap's relative half-width exceeds 0.25, before the simulation is read (sensitivity). Rows of `12_verdicts.csv` with analysis `headline`.

Table S20

Level 2 per Model, Marginal Estimand

Contrast	Model	Pop. gap	Sim. gap	Ratio	98.6% interval	Verdict	Population-only stop
Black minus White	DeepSeek-V3	0.33	0.21	0.65	[−0.31, 2.32]	reference too imprecise	reference too imprecise
	Gemini-3-Flash		−0.39	−1.20	[−4.24, 0.52]	undetermined	reference too imprecise
	GPT-4o-mini		0.22	0.66	[−0.16, 2.21]	reference too imprecise	reference too imprecise
	GLM-4.7		−0.56	−1.69	[−5.15, −0.25]	reversed	reference too imprecise
	Pooled		−0.13	−0.40	[−1.46, 0.18]	reversed or missing	reference too imprecise
Hispanic minus White	DeepSeek-V3	0.24	0.17	0.72	[−0.79, 3.85]	reference too imprecise	reference too imprecise
	Gemini-3-Flash		−0.01	−0.03	[−1.54, 1.42]	undetermined	reference too imprecise
	GPT-4o-mini		0.14	0.58	[−0.13, 2.65]	reference too imprecise	reference too imprecise
	GLM-4.7		−0.44	−1.83	[−8.15, 0.15]	reversed or missing	reference too imprecise
	Pooled		−0.03	−0.14	[−1.24, 0.62]	undetermined	reference too imprecise
Asian minus White	DeepSeek-V3	−0.76	−0.71	0.94	[0.45, 1.53]	reference too imprecise	reference too imprecise
	Gemini-3-Flash		−1.45	1.92	[−0.50, 4.49]	undetermined	reference too imprecise
	GPT-4o-mini		−0.40	0.53	[0.12, 1.00]	undetermined	reference too imprecise
	GLM-4.7		−1.74	2.30	[0.69, 4.14]	undetermined	reference too imprecise
	Pooled		−1.07	1.42	[0.72, 2.31]	undetermined	reference too imprecise
Women minus Men	DeepSeek-V3	0.94	0.35	0.37	[0.20, 0.54]	missing or attenuated	missing or attenuated
	Gemini-3-Flash		0.99	1.05	[0.66, 1.47]	undetermined	undetermined
	GPT-4o-mini		0.19	0.20	[0.00, 0.41]	missing or attenuated	missing or attenuated
	GLM-4.7		1.73	1.84	[1.38, 2.36]	steepened	steepened

Table S20, continued

Contrast	Model	Pop. gap	Sim. gap	Ratio	98.6% interval	Verdict	Population-only stop
Low minus High SES	Pooled		0.81	0.87	[0.69, 1.07]	attenuated or kept	attenuated or kept
	DeepSeek-V3	2.02	2.48	1.23	[1.04, 1.44]	kept or steepened	kept or steepened
	Gemini-3-Flash		6.58	3.26	[2.54, 4.03]	steepened	steepened
	GPT-4o-mini		2.80	1.39	[1.21, 1.59]	kept or steepened	kept or steepened
	GLM-4.7		6.53	3.23	[2.71, 3.81]	steepened	steepened
Middle minus High SES	Pooled		4.60	2.28	[2.00, 2.60]	steepened	steepened
	DeepSeek-V3	0.89	1.56	1.75	[1.37, 2.21]	steepened	steepened
	Gemini-3-Flash		0.79	0.89	[0.63, 1.19]	attenuated or kept	attenuated or kept
	GPT-4o-mini		1.37	1.53	[1.17, 1.98]	kept or steepened	kept or steepened
	GLM-4.7		1.34	1.50	[0.94, 2.13]	kept or steepened	kept or steepened
Low minus Middle SES	Pooled		1.26	1.42	[1.17, 1.75]	kept or steepened	kept or steepened
	DeepSeek-V3	1.13	0.92	0.82	[0.53, 1.16]	attenuated or kept	attenuated or kept
	Gemini-3-Flash		5.79	5.13	[3.81, 6.83]	steepened	steepened
	GPT-4o-mini		1.44	1.27	[1.02, 1.62]	kept or steepened	kept or steepened
	GLM-4.7		5.19	4.60	[3.72, 5.83]	steepened	steepened
	Pooled		3.33	2.95	[2.40, 3.74]	steepened	steepened

Note. Gaps in PHQ-8 points. Ratio = simulated gap over population gap, with its Fieller interval at the Bonferroni level for 7 contrasts (one-sided $\alpha = .0071$). Verdict: the primary rule, whose reference-imprecision stop fires when the observed simulated gap, treated as measured without error, would still give a ratio interval spanning three or more regions. Population-only stop: the verdict when that stop instead fires whenever the population gap's relative half-width exceeds 0.25, before the simulation is read (sensitivity). Rows of `12_verdicts.csv` with analysis `headline`.

S4.3 Level 3

Table S21 gives every group row for the mean, and Table S22 every group row for the share scoring 10 or more.

Table S21

Level 3 per Model and Group: Mean PHQ-8

Model	Group	Sim.	Ref.	Residual	90% CI	Passes at tolerance			
						0.2 SD	1 pt	2 pt	5 pt (bench.)
DeepSeek-V3	Overall	7.48	2.89	4.58	[4.49, 4.67]	No	No	No	Yes
	White	7.28	2.86	4.42	[4.31, 4.54]	No	No	No	Yes
	Black	8.26	3.20	5.05	[4.91, 5.20]	No	No	No	No
	Asian	6.63	2.16	4.47	[4.32, 4.61]	No	No	No	Yes
	Hispanic	8.22	3.13	5.09	[4.96, 5.22]	No	No	No	No
	Men	7.23	2.43	4.80	[4.69, 4.92]	No	No	No	Yes
	Women	7.71	3.34	4.38	[4.26, 4.49]	No	No	No	Yes
	Low	9.27	4.14	5.13	[4.96, 5.30]	No	No	No	No
	Middle	7.99	3.07	4.92	[4.82, 5.02]	No	No	No	No
	High	6.24	2.18	4.06	[3.91, 4.20]	No	No	No	Yes
Gemini-3-Flash	Overall	5.48	2.89	2.59	[2.49, 2.69]	No	No	No	Yes
	White	5.09	2.86	2.23	[2.12, 2.35]	No	No	No	Yes
	Black	6.55	3.20	3.34	[3.16, 3.53]	No	No	No	Yes
	Asian	4.58	2.16	2.42	[2.26, 2.58]	No	No	No	Yes

Table S21, continued

Model	Group	Sim.	Ref.	Residual	90% CI	Passes at tolerance			
						0.2 SD	1 pt	2 pt	5 pt (bench.)
	Hispanic	7.09	3.13	3.97	[3.77, 4.16]	No	No	No	Yes
	Men	4.91	2.43	2.48	[2.37, 2.59]	No	No	No	Yes
	Women	6.03	3.34	2.69	[2.56, 2.82]	No	No	No	Yes
	Low	11.51	4.14	7.37	[7.20, 7.53]	No	No	No	No
	Middle	4.40	3.07	1.33	[1.22, 1.44]	No	No	Yes	Yes
	High	3.60	2.18	1.42	[1.32, 1.52]	No	No	Yes	Yes
GPT-4o-mini	Overall	7.57	2.89	4.68	[4.57, 4.79]	No	No	No	Yes
	White	7.37	2.86	4.51	[4.36, 4.66]	No	No	No	Yes
	Black	8.27	3.20	5.07	[4.90, 5.24]	No	No	No	No
	Asian	7.15	2.16	4.98	[4.81, 5.15]	No	No	No	No
	Hispanic	8.30	3.13	5.17	[5.02, 5.33]	No	No	No	No
	Men	7.34	2.43	4.91	[4.77, 5.05]	No	No	No	No
	Women	7.80	3.34	4.46	[4.30, 4.62]	No	No	No	Yes
	Low	9.46	4.14	5.32	[5.13, 5.51]	No	No	No	No
	Middle	7.95	3.07	4.88	[4.73, 5.02]	No	No	No	No
	High	6.41	2.18	4.23	[4.04, 4.42]	No	No	No	Yes
GLM-4.7	Overall	5.02	2.89	2.12	[2.00, 2.24]	No	No	No	Yes
	White	4.72	2.86	1.86	[1.71, 2.01]	No	No	No	Yes
	Black	6.16	3.20	2.96	[2.75, 3.17]	No	No	No	Yes
	Asian	3.66	2.16	1.50	[1.30, 1.69]	No	No	Yes	Yes
	Hispanic	6.20	3.13	3.08	[2.89, 3.27]	No	No	No	Yes
	Men	3.94	2.43	1.51	[1.39, 1.63]	No	No	Yes	Yes
	Women	6.03	3.34	2.69	[2.52, 2.86]	No	No	No	Yes
	Low	10.54	4.14	6.40	[6.21, 6.59]	No	No	No	No
	Middle	4.57	3.07	1.50	[1.34, 1.66]	No	No	Yes	Yes
	High	2.84	2.18	0.66	[0.51, 0.81]	No	Yes	Yes	Yes
Pooled	Overall	6.39	2.89	3.49	[3.42, 3.57]	No	No	No	Yes
	White	6.12	2.86	3.26	[3.16, 3.36]	No	No	No	Yes
	Black	7.31	3.20	4.11	[3.96, 4.25]	No	No	No	Yes
	Asian	5.51	2.16	3.34	[3.20, 3.48]	No	No	No	Yes
	Hispanic	7.45	3.13	4.33	[4.19, 4.47]	No	No	No	Yes
	Men	5.85	2.43	3.43	[3.34, 3.52]	No	No	No	Yes
	Women	6.89	3.34	3.55	[3.45, 3.66]	No	No	No	Yes
	Low	10.19	4.14	6.05	[5.91, 6.19]	No	No	No	No
	Middle	6.23	3.07	3.16	[3.06, 3.25]	No	No	No	Yes
	High	4.77	2.18	2.59	[2.50, 2.69]	No	No	No	Yes

Note. Post-stratified estimand, specification S1, framings combined. Residual = simulated minus reference, PHQ-8 points. 0.2 SD = 0.79 points. A row passes when its 90% interval lies inside the tolerance (two one-sided tests). The 5-point column is a benchmark only.

Table S22*Level 3 per Model and Group: Share Scoring 10 or More*

Model	Group	Sim. %	Ref. %	Diff. (pp)	90% CI	Ratio	90% CI	Passes		
								2.5 pp	5 pp	0.80–1.25
DeepSeek-V3	Overall	7.4	7.0	0.3	[−1.0, 1.7]	1.05	[0.85, 1.25]	Yes	Yes	Yes
	White	6.1	6.8	−0.8	[−2.6, 1.1]	0.89	[0.62, 1.17]	No	Yes	No
	Black	14.0	8.8	5.1	[2.7, 7.6]	1.58	[1.30, 1.88]	No	No	No
	Asian	1.8	3.4	−1.6	[−3.4, 0.1]	0.53	[0.04, 1.05]	No	Yes	No
	Hispanic	11.2	8.1	3.1	[0.9, 5.3]	1.38	[1.10, 1.67]	No	No	No
	Men	4.8	5.2	−0.4	[−2.3, 1.6]	0.93	[0.56, 1.32]	Yes	Yes	No
	Women	9.7	8.7	1.0	[−0.9, 2.9]	1.11	[0.89, 1.34]	No	Yes	No
	Low	32.7	13.6	19.1	[16.0, 22.1]	2.40	[2.15, 2.68]	No	No	No
	Middle	1.4	7.3	−6.0	[−7.9, −4.0]	0.19	[0.00, 0.44]	No	No	No
	High	0.5	3.7	−3.2	[−5.5, −0.9]	0.14	[0.00, 0.75]	No	No	No
Gemini-3-Flash	Overall	16.4	7.0	9.4	[7.9, 10.9]	2.34	[2.12, 2.57]	No	No	No
	White	13.2	6.8	6.4	[4.5, 8.3]	1.94	[1.65, 2.24]	No	No	No
	Black	26.3	8.8	17.5	[15.0, 19.9]	2.98	[2.68, 3.30]	No	No	No
	Asian	4.4	3.4	1.0	[−0.9, 2.9]	1.30	[0.75, 1.93]	No	Yes	No
	Hispanic	30.7	8.1	22.6	[20.2, 25.1]	3.79	[3.41, 4.22]	No	No	No
	Men	12.8	5.2	7.6	[5.7, 9.5]	2.47	[2.07, 2.89]	No	No	No
	Women	19.8	8.7	11.1	[9.1, 13.1]	2.27	[2.03, 2.52]	No	No	No
	Low	81.1	13.6	67.5	[65.2, 69.9]	5.96	[5.57, 6.39]	No	No	No
	Middle	0.0	7.3	−7.3	[−9.1, −5.5]	0.00	[0.00, 0.23]	No	No	No
	High	0.0	3.7	−3.7	[−5.9, −1.6]	0.00	[0.00, 0.57]	No	No	No
GPT-4o-mini	Overall	16.9	7.0	9.9	[8.2, 11.7]	2.41	[2.15, 2.69]	No	No	No
	White	14.5	6.8	7.7	[5.3, 10.0]	2.13	[1.77, 2.50]	No	No	No
	Black	26.5	8.8	17.7	[14.8, 20.6]	3.00	[2.64, 3.38]	No	No	No
	Asian	11.2	3.4	7.8	[5.4, 10.2]	3.31	[2.50, 4.31]	No	No	No
	Hispanic	24.5	8.1	16.4	[13.7, 19.2]	3.03	[2.64, 3.45]	No	No	No
	Men	14.0	5.2	8.9	[6.6, 11.1]	2.71	[2.25, 3.20]	No	No	No
	Women	19.6	8.7	10.9	[8.3, 13.5]	2.25	[1.94, 2.57]	No	No	No
	Low	46.6	13.6	33.0	[29.8, 36.2]	3.42	[3.12, 3.76]	No	No	No
	Middle	14.4	7.3	7.1	[4.3, 9.9]	1.96	[1.57, 2.37]	No	No	No
	High	5.3	3.7	1.6	[−1.2, 4.4]	1.44	[0.69, 2.22]	No	Yes	No
GLM-4.7	Overall	13.2	7.0	6.2	[4.8, 7.7]	1.89	[1.68, 2.11]	No	No	No
	White	11.4	6.8	4.6	[2.7, 6.5]	1.68	[1.40, 1.98]	No	No	No
	Black	20.8	8.8	11.9	[9.4, 14.4]	2.35	[2.05, 2.66]	No	No	No
	Asian	3.6	3.4	0.2	[−1.7, 2.1]	1.06	[0.53, 1.66]	Yes	Yes	No
	Hispanic	20.7	8.1	12.6	[10.2, 14.9]	2.55	[2.23, 2.91]	No	No	No
	Men	7.8	5.2	2.6	[0.7, 4.5]	1.51	[1.14, 1.89]	No	Yes	No
	Women	18.3	8.7	9.6	[7.6, 11.6]	2.10	[1.86, 2.35]	No	No	No
	Low	61.4	13.6	47.8	[44.9, 50.6]	4.51	[4.18, 4.87]	No	No	No
	Middle	2.0	7.3	−5.4	[−7.4, −3.4]	0.27	[0.01, 0.53]	No	No	No
	High	0.3	3.7	−3.5	[−5.6, −1.3]	0.07	[0.00, 0.65]	No	No	No
Pooled	Overall	13.5	7.0	6.5	[5.6, 7.4]	1.92	[1.78, 2.07]	No	No	No
	White	11.3	6.8	4.5	[3.3, 5.7]	1.66	[1.48, 1.85]	No	No	No
	Black	21.9	8.8	13.1	[11.4, 14.7]	2.48	[2.27, 2.70]	No	No	No
	Asian	5.3	3.4	1.9	[0.6, 3.1]	1.55	[1.17, 2.02]	No	Yes	No
	Hispanic	21.8	8.1	13.7	[12.2, 15.2]	2.69	[2.44, 2.97]	No	No	No
	Men	9.9	5.2	4.7	[3.6, 5.8]	1.90	[1.67, 2.16]	No	No	No
	Women	16.9	8.7	8.1	[6.9, 9.4]	1.93	[1.77, 2.10]	No	No	No
	Low	55.5	13.6	41.9	[40.2, 43.5]	4.07	[3.81, 4.37]	No	No	No
	Middle	4.4	7.3	−2.9	[−4.1, −1.7]	0.61	[0.46, 0.76]	No	Yes	No
	High	1.5	3.7	−2.2	[−3.4, −0.9]	0.41	[0.10, 0.74]	No	Yes	No

Note. Post-stratified estimand, specification S1, framings combined. pp = percentage points. Ratio = simulated share over reference share, Fieller interval.

S4.4 Level 4 Invariance Steps

Level 4 Invariance Steps: RMSEA of the Difference

Population	Attribute	Step	G	d	RMSEA $_D$	90% CI	$\hat{c}$	At .08	At .05
NHANES 2005-2018 (weighted)	sex	metric	2	7	.033	[.020, .044]	4.18	holds	holds
	sex	scalar	2	7	.067	[.061, .073]	1.67	holds	fails
	sex	metric (polychoric)	2	7	.020	[.009, .029]	3.14	holds	holds
	race	metric	4	21	.033	[.019, .043]	5.20	holds	holds
	race	scalar	4	21	.052	[.048, .056]	3.24	holds	indeterminate
	race	metric (polychoric)	4	21	.037	[.027, .047]	3.42	holds	holds
	income	metric	3	14	.051	[.037, .062]	4.58	holds	indeterminate
	income	scalar	3	14	.026	[.018, .032]	1.63	holds	holds
	income	metric (polychoric)	3	14	.031	[.021, .042]	3.16	holds	holds
DeepSeek-V3 clinical	sex	metric	2	7	.100	–	–	not interpretable (no general factor)	not interpretable (no general factor)
	sex	scalar	2	7	.059	–	–	not interpretable (no general factor)	not interpretable (no general factor)
	sex	metric (polychoric)	2	7	.052	–	–	not interpretable (no general factor)	not interpretable (no general factor)
	race	metric	4	21	.109	–	–	not interpretable (no general factor)	not interpretable (no general factor)
	race	scalar	4	21	.072	–	–	not interpretable (no general factor)	not interpretable (no general factor)
	race	metric (polychoric)	4	21	.145	–	–	not interpretable (no general factor)	not interpretable (no general factor)
	income	metric	3	14	.205	–	–	not interpretable (no general factor)	not interpretable (no general factor)
	income	scalar	3	14	.223	–	–	not interpretable (no general factor)	not interpretable (no general factor)
	income	metric (polychoric)	3	14	.243	–	–	not interpretable (no general factor)	not interpretable (no general factor)
DeepSeek-V3 narrative	sex	metric	2	7	.033	–	–	not interpretable (no general factor)	not interpretable (no general factor)
	sex	scalar	2	7	.047	–	–	not interpretable (no general factor)	not interpretable (no general factor)
	sex	metric (polychoric)	2	7	.330	–	–	not interpretable (no general factor)	not interpretable (no general factor)
	race	metric	4	21	.139	–	–	not interpretable (no general factor)	not interpretable (no general factor)
	race	scalar	4	21	.152	–	–	not interpretable (no general factor)	not interpretable (no general factor)
	race	metric (polychoric)	4	21	.144	–	–	not interpretable (no general factor)	not interpretable (no general factor)
	income	metric	3	14	.211	–	–	not interpretable (no general factor)	not interpretable (no general factor)

Table S23, continued

Population	Attribute	Step	<i>G</i>	<i>d</i>	RMSEA _D	90% CI	$\hat{c}$	At .08	At .05
	income	scalar	3	14	.263	–	–	not interpretable (no general factor)	not interpretable (no general factor)
	income	metric (polychoric)	3	14	.252	–	–	not interpretable (no general factor)	not interpretable (no general factor)
Gemini-3-Flash clinical	sex	metric	2	7	.070	[.000, .127]	4.79	indeterminate	indeterminate
	sex	scalar	2	7	.116	[.052, .183]	4.83	indeterminate	fails
	sex	metric (polychoric)	2	7	.047	[.000, .138]	7.39	indeterminate	indeterminate
	race	metric	4	21	.000	[.000, .062]	4.39	holds	indeterminate
	race	scalar	4	21	.040	[.000, .104]	5.28	indeterminate	indeterminate
	race	metric (polychoric)	4	21	.000	[.000, .080]	5.80	holds	indeterminate
	income	metric	3	14	.192	[.151, .230]	4.19	fails	fails
	income	scalar	3	14	.096	[.061, .125]	5.49	indeterminate	fails
	income	metric (polychoric)	3	14	.148	[.053, .197]	5.74	indeterminate	fails
Gemini-3-Flash narrative	sex	metric	2	7	.039	[.000, .105]	2.85	indeterminate	indeterminate
	sex	scalar	2	7	.034	[.000, .120]	4.38	indeterminate	indeterminate
	sex	metric (polychoric)	2	7	.000	[.000, .141]	13.66	indeterminate	indeterminate
	race	metric	4	21	.000	[.000, .037]	3.88	holds	holds
	race	scalar	4	21	.000	[.000, .073]	4.23	holds	indeterminate
	race	metric (polychoric)	4	21	.069	[.000, .127]	4.84	indeterminate	indeterminate
	income	metric	3	14	.159	[.134, .186]	4.00	fails	fails
	income	scalar	3	14	.160	[.123, .196]	4.17	fails	fails
	income	metric (polychoric)	3	14	.167	[.120, .216]	5.09	fails	fails
GLM-4.7 clinical	sex	metric	2	7	.000	[.000, .072]	2.57	holds	indeterminate
	sex	scalar	2	7	.081	[.048, .108]	2.17	indeterminate	indeterminate
	sex	metric (polychoric)	2	7	.000	[.000, .059]	2.67	holds	indeterminate
	race	metric	4	21	.035	[.000, .071]	2.16	holds	indeterminate
	race	scalar	4	21	.043	[.000, .078]	2.82	holds	indeterminate
	race	metric (polychoric)	4	21	.012	[.000, .062]	2.11	holds	indeterminate
	income	metric	3	14	.162	[.139, .185]	2.28	fails	fails
	income	scalar	3	14	.160	[.136, .184]	2.87	fails	fails
	income	metric (polychoric)	3	14	.081	[.053, .105]	2.14	indeterminate	fails
GLM-4.7 narrative	sex	metric	2	7	.102	[.023, .149]	3.00	indeterminate	indeterminate
	sex	scalar	2	7	.052	[.000, .091]	2.13	indeterminate	indeterminate
	sex	metric (polychoric)	2	7	.044	[.000, .108]	3.22	indeterminate	indeterminate
	race	metric	4	21	.000	[.000, .053]	2.66	holds	indeterminate
	race	scalar	4	21	.000	[.000, .060]	3.26	holds	indeterminate
	race	metric (polychoric)	4	21	.000	[.000, .031]	2.77	holds	holds
	income	metric	3	14	.199	[.180, .219]	2.70	fails	fails
	income	scalar	3	14	.190	[.165, .212]	3.42	fails	fails
	income	metric (polychoric)	3	14	.123	[.097, .148]	2.18	fails	fails
GPT-4o-mini clinical	sex	metric	2	7	.000	–	–	not interpretable (no general factor)	not interpretable (no general factor)

Table S23, continued

Population	Attribute	Step	G	d	RMSEA _D	90% CI	$\hat{c}$	At .08	At .05
	sex	scalar	2	7	.033	–	–	not interpretable (no general factor)	not interpretable (no general factor)
	sex	metric (polychoric)	2	7	.022	–	–	not interpretable (no general factor)	not interpretable (no general factor)
	race	metric	4	21	.019	–	–	not interpretable (no general factor)	not interpretable (no general factor)
	race	scalar	4	21	.035	–	–	not interpretable (no general factor)	not interpretable (no general factor)
	race	metric (polychoric)	4	21	.076	–	–	not interpretable (no general factor)	not interpretable (no general factor)
	income	metric	3	14	.121	–	–	not interpretable (no general factor)	not interpretable (no general factor)
	income	scalar	3	14	.246	–	–	not interpretable (no general factor)	not interpretable (no general factor)
	income	metric (polychoric)	3	14	.153	–	–	not interpretable (no general factor)	not interpretable (no general factor)
GPT-4o-mini narrative	sex	metric	2	7	.006	–	–	not interpretable (no general factor)	not interpretable (no general factor)
	sex	scalar	2	7	.031	–	–	not interpretable (no general factor)	not interpretable (no general factor)
	sex	metric (polychoric)	2	7	.028	–	–	not interpretable (no general factor)	not interpretable (no general factor)
	race	metric	4	21	.048	–	–	not interpretable (no general factor)	not interpretable (no general factor)
	race	scalar	4	21	.060	–	–	not interpretable (no general factor)	not interpretable (no general factor)
	race	metric (polychoric)	4	21	.066	–	–	not interpretable (no general factor)	not interpretable (no general factor)
	income	metric	3	14	.099	–	–	not interpretable (no general factor)	not interpretable (no general factor)
	income	scalar	3	14	.169	–	–	not interpretable (no general factor)	not interpretable (no general factor)
	income	metric (polychoric)	3	14	.106	–	–	not interpretable (no general factor)	not interpretable (no general factor)

Note. A step holds when the upper 90% limit of RMSEA_D is below the margin, fails when the lower limit is above it, and is indeterminate otherwise. The margin of .08 is primary and .05 a sensitivity analysis. $\hat{c}$ = permutation scaling of the expected ΔF under exact invariance. The polychoric metric rows are reported as fitted.

S4.5 Multiplicity

Table S24 gives the family and the procedure at each rung. Each rung's pass is an intersection-union test over its members, except level 2, where each contrast receives its

own verdict and the family is controlled by Bonferroni.

Table S24

Multiplicity: Family and Procedure at Each Rung

Rung	Claim	Family	Members	Procedure
Gate	Draws needed per persona	Each model $\times$ framing (4×2)	Estimation of $\phi(k)$ and $SE(k)$; no test	None; persona-bootstrap intervals reported
Level 1	Person-fit like people's	Each model $\times$ framing	2 tails: misfit ratio and overfit ratio, each against $[1/\tau, \tau]$, $\tau = 1.5$	Intersection-union: pass needs both 90% intervals inside the band; no adjustment
Level 2	Group gaps like people's	Each model (the pooled row is description)	7 contrasts	Bonferroni: one-sided $\alpha = .05/7 = .0071$, 98.6% Fieller intervals
Level 3	Group levels like people's, at a stated tolerance	Each model $\times$ tolerance	10 group rows	Intersection-union: pass needs every row's 90% interval inside the tolerance (two one-sided tests at .05); no adjustment
Level 4	Structure like people's	Each model	R1, R2 and R4 in each of 2 framings; R4 reads 6 invariance steps	Intersection-union: pass needs every rule in every framing; R4 needs every step to match the reference's verdict; R3 is a diagnostic and not counted

Note. Family sizes, τ , α and the interval level are read from the receipts named in the source comment of this table.

S5. Controls

Each replicate splits the NHANES 2005–2018 adults at random into two halves, stratified by the 48 persona cells. One half supplies respondents and the other half is the reference. A pseudo-model has the corpus design of 120 personas, two framings and 30 draws. Each draw is a respondent from the donor half, drawn within the persona's cell with probability proportional to the examination weight, and it keeps that respondent's eight item scores. Transgender and multiracial personas draw from the closest available group and enter levels 1 and 4 only. Four independent pseudo-models form a panel, and there are 200 replicates. Planted failures are applied to the same panels at a grid of doses. At level 1 a share of draws has its item values shuffled, or is replaced by the most typical vector at the same total. At level 2 a group gap is scaled by a factor g . At level 3 every persona mean is

shifted by d points. At level 4 a share of draws is rebuilt item by item from other draws of the same persona. Table S28 adds a parametric simulation at the corpus's own precision, with the simulated-gap standard error set to the smallest and the largest of the four real models. The design and scripts are described in `paper_brm/analysis_brm/CONTROLS.md`.

Table S25*Controls, Level 1: Detection by Dose*

Condition	Dose (%)	Unit	n	Expected reading (%)	Pass	Compressed	Excess misfit	Undet.	Misfit	Overfit
Positive control	0.0	both	800	99.4 [98.5, 99.7]	99.4	0.0	0.0	0.6	1.15	1.00
	0.0	clinical	800	97.6 [96.3, 98.5]	97.6	0.0	0.0	2.4	1.15	1.00
	0.0	narrative	800	96.4 [94.8, 97.5]	96.4	0.0	0.0	3.6	1.15	1.01
Shuffle item values	10.0	both	200	0.0 [0.0, 1.9]	28.5	0.0	0.0	71.5	1.39	0.95
	10.0	clinical	200	0.5 [0.1, 2.8]	21.5	0.0	0.5	78.0	1.39	0.95
	10.0	narrative	200	0.5 [0.1, 2.8]	24.5	0.0	0.5	75.0	1.38	0.95
	20.0	both	200	49.5 [42.6, 56.4]	0.0	0.0	49.5	50.5	1.63	0.90
	20.0	clinical	200	42.5 [35.9, 49.4]	0.0	0.0	42.5	57.5	1.64	0.89
	20.0	narrative	200	35.5 [29.2, 42.3]	0.0	0.0	35.5	64.5	1.62	0.90
	40.0	both	200	100.0 [98.1, 100.0]	0.0	0.0	100.0	0.0	2.11	0.79
	40.0	clinical	200	100.0 [98.1, 100.0]	0.0	0.0	100.0	0.0	2.11	0.79
	40.0	narrative	200	100.0 [98.1, 100.0]	0.0	0.0	100.0	0.0	2.11	0.79
	60.0	both	200	100.0 [98.1, 100.0]	0.0	0.0	100.0	0.0	2.59	0.68
	60.0	clinical	200	100.0 [98.1, 100.0]	0.0	0.0	100.0	0.0	2.59	0.68
	60.0	narrative	200	100.0 [98.1, 100.0]	0.0	0.0	100.0	0.0	2.58	0.68
	80.0	both	200	100.0 [98.1, 100.0]	0.0	0.0	100.0	0.0	3.07	0.58
	80.0	clinical	200	100.0 [98.1, 100.0]	0.0	0.0	100.0	0.0	3.08	0.58
	80.0	narrative	200	100.0 [98.1, 100.0]	0.0	0.0	100.0	0.0	3.06	0.58
	100.0	both	200	100.0 [98.1, 100.0]	0.0	0.0	100.0	0.0	3.55	0.47
	100.0	clinical	200	100.0 [98.1, 100.0]	0.0	0.0	100.0	0.0	3.55	0.47
	100.0	narrative	200	100.0 [98.1, 100.0]	0.0	0.0	100.0	0.0	3.54	0.47
Most typical vector	10.0	both	200	0.0 [0.0, 1.9]	100.0	0.0	0.0	0.0	1.06	0.94
	10.0	clinical	200	0.0 [0.0, 1.9]	100.0	0.0	0.0	0.0	1.07	0.94
	10.0	narrative	200	0.0 [0.0, 1.9]	100.0	0.0	0.0	0.0	1.06	0.94
	20.0	both	200	0.0 [0.0, 1.9]	100.0	0.0	0.0	0.0	0.98	0.88
	20.0	clinical	200	0.0 [0.0, 1.9]	100.0	0.0	0.0	0.0	0.98	0.88
	20.0	narrative	200	0.0 [0.0, 1.9]	99.5	0.0	0.0	0.5	0.98	0.88
	40.0	both	200	0.0 [0.0, 1.9]	72.5	0.0	0.0	27.5	0.82	0.76
	40.0	clinical	200	0.0 [0.0, 1.9]	54.0	0.0	0.0	46.0	0.82	0.76
	40.0	narrative	200	0.0 [0.0, 1.9]	52.5	0.0	0.0	47.5	0.82	0.76
	60.0	both	200	60.0 [53.1, 66.5]	0.0	2.0	0.0	40.0	0.66	0.63
	60.0	clinical	200	48.0 [41.2, 54.9]	0.0	0.5	0.0	52.0	0.65	0.63
	60.0	narrative	200	44.5 [37.8, 51.4]	0.0	1.0	0.0	55.5	0.66	0.64
	80.0	both	200	100.0 [98.1, 100.0]	0.0	80.0	0.0	0.0	0.49	0.51
	80.0	clinical	200	100.0 [98.1, 100.0]	0.0	80.0	0.0	0.0	0.49	0.51
	80.0	narrative	200	100.0 [98.1, 100.0]	0.0	80.0	0.0	0.0	0.49	0.51
	100.0	both	200	100.0 [98.1, 100.0]	0.0	89.5	0.0	0.0	0.33	0.39
	100.0	clinical	200	100.0 [98.1, 100.0]	0.0	86.5	0.0	0.0	0.33	0.39
	100.0	narrative	200	100.0 [98.1, 100.0]	0.0	88.5	0.0	0.0	0.33	0.39

Note. Dose = share of draws altered. Unit: **both** = a pseudo-model read on both framings; **clinical** and **narrative** = one framing. Expected reading = share of replicates returning the reading the planted failure should produce (a pass for the positive control), with a 95% interval. Pass, Compressed, Excess misfit and Undet. are shares of replicates in percent. Misfit and Overfit are mean ratios.

Table S26*Controls, Level 2: Verdicts by Dose*

Reference, scope	Contrast	<i>g</i>	True region	<i>n</i>	Correct	Correct or compatible	Wrong	Undet.	Stopped
half sample, per model	Black minus White	−0.50	reversed	800	0.9	2.5	0.0	40.5	57.0
		0.00	missing	800	0.0	1.0	0.0	32.1	66.9
		0.50	attenuated	800	0.0	0.0	0.4	24.4	75.2
		1.00	kept	800	0.0	0.0	0.0	18.9	81.1
		1.50	steepened	800	0.0	0.5	0.0	22.0	77.5
		2.00	steepened	800	0.1	1.4	0.0	28.9	69.8
	Hispanic minus White	3.00	steepened	800	3.6	13.6	0.0	32.4	54.0
		1.00	kept	800	0.0	0.0	0.0	8.8	91.2
	Asian minus White	1.00	kept	800	0.0	0.8	0.0	73.8	25.5
	Women minus Men	−0.50	reversed	800	4.4	75.0	0.0	25.0	0.0
		0.00	missing	800	0.0	5.0	0.0	95.0	0.0
		0.50	attenuated	800	0.0	1.5	0.0	95.1	3.4
		1.00	kept	800	0.0	5.1	0.0	88.1	6.8
		1.50	steepened	800	4.1	75.0	0.0	25.0	0.0
		2.00	steepened	800	65.5	99.9	0.0	0.1	0.0
	Low minus High SES	3.00	steepened	800	100.0	100.0	0.0	0.0	0.0
		−0.50	reversed	800	3.1	81.5	0.0	18.5	0.0
		0.00	missing	800	0.0	4.0	0.0	96.0	0.0
		0.50	attenuated	800	0.0	2.8	0.0	97.0	0.2
		1.00	kept	800	0.0	5.4	0.0	93.6	1.0
		1.50	steepened	800	3.4	84.6	0.0	15.4	0.0
	Middle minus High SES	2.00	steepened	800	78.5	100.0	0.0	0.0	0.0
		3.00	steepened	800	100.0	100.0	0.0	0.0	0.0
		−0.50	reversed	800	2.6	38.8	0.0	60.9	0.4
		0.00	missing	800	0.0	3.2	0.0	92.8	4.0
		0.50	attenuated	800	0.0	0.2	0.1	72.0	27.6
		1.00	kept	800	0.0	4.0	0.1	56.4	39.5
	Low minus Middle SES	1.50	steepened	800	3.5	35.1	0.0	56.1	8.8
		2.00	steepened	800	28.2	82.9	0.0	16.8	0.4
		3.00	steepened	800	97.4	100.0	0.0	0.0	0.0
		1.00	kept	800	0.0	2.2	0.1	64.9	32.8
half sample, pooled	Black minus White	−0.50	reversed	200	2.5	19.5	0.0	29.5	51.0
		0.00	missing	200	0.0	5.0	0.0	32.5	62.5
		0.50	attenuated	200	0.0	0.0	0.0	20.5	79.5
		1.00	kept	200	0.0	0.0	0.0	8.5	91.5
		1.50	steepened	200	0.0	1.0	0.0	13.5	85.5
		2.00	steepened	200	0.0	11.0	0.0	21.0	68.0
	Hispanic minus White	3.00	steepened	200	14.5	43.5	0.0	6.0	50.5
		1.00	kept	200	0.0	0.0	0.0	5.5	94.5
	Asian minus White	1.00	kept	200	0.0	19.0	0.0	57.0	24.0
	Women minus Men	−0.50	reversed	200	29.5	100.0	0.0	0.0	0.0
		0.00	missing	200	0.0	70.5	0.0	29.5	0.0
		0.50	attenuated	200	0.0	62.0	0.0	36.5	1.5
		1.00	kept	200	0.0	45.5	0.0	45.5	9.0
		1.50	steepened	200	17.0	100.0	0.0	0.0	0.0
		2.00	steepened	200	96.5	100.0	0.0	0.0	0.0
	Low minus High SES	3.00	steepened	200	100.0	100.0	0.0	0.0	0.0
		−0.50	reversed	200	41.0	100.0	0.0	0.0	0.0
		0.00	missing	200	6.0	83.0	0.0	17.0	0.0
		0.50	attenuated	200	0.0	70.5	0.0	29.5	0.0
		1.00	kept	200	0.0	49.5	0.0	50.0	0.5
		1.50	steepened	200	18.0	100.0	0.0	0.0	0.0
		2.00	steepened	200	99.5	100.0	0.0	0.0	0.0
		3.00	steepened	200	100.0	100.0	0.0	0.0	0.0

Table S26, continued

Reference, scope	Contrast	<i>g</i>	True region	<i>n</i>	Correct	Correct or compatible	Wrong	Undet.	Stopped
	Low minus High SES	3.00	steepened	200	100.0		100.0	0.0	0.0
		−0.50	reversed	200	43.0		100.0	0.0	0.0
		0.00	missing	200	6.5		85.0	0.0	15.0
		0.50	attenuated	200	1.5		82.0	0.0	18.0
		1.00	kept	200	0.0		68.0	0.0	32.0
		1.50	steepened	200	34.0		100.0	0.0	0.0
	Middle minus High SES	2.00	steepened	200	100.0		100.0	0.0	0.0
		3.00	steepened	200	100.0		100.0	0.0	0.0
		−0.50	reversed	200	20.5		96.5	0.0	3.5
		0.00	missing	200	0.0		32.5	0.0	67.0
		0.50	attenuated	200	0.0		24.5	1.0	63.5
		1.00	kept	200	0.0		32.5	1.0	53.5
	Low minus Middle SES	1.50	steepened	200	17.5		86.5	0.0	13.0
		2.00	steepened	200	75.5		100.0	0.0	0.0
		3.00	steepened	200	100.0		100.0	0.0	0.0
		1.00	kept	200	0.0		23.5	0.5	58.0

Note. *g* = planted ratio of the simulated gap to the population gap. Shares in percent. Correct = the single true region named; Correct or compatible = a verdict that contains the true region; Wrong = a determinate verdict that excludes it; Stopped = either stop. Half sample: the reference is the other NHANES half. Audit precision: the reference standard error is scaled to the full NHANES sample.

Table S27

Controls, Level 2: Coverage of the True Ratio

Scope	Contrast	<i>g</i>	<i>n</i>	Coverage	Mean ratio	SD of ratio
per model	Black minus White	−0.50	800	.995	−0.44	0.90
		0.00	800	.995	0.10	0.96
		0.50	800	.993	0.65	1.06
		1.00	800	.991	1.19	1.20
		1.50	800	.993	1.74	1.36
		2.00	800	.996	2.28	1.54
		3.00	800	.999	3.37	1.93
	Hispanic minus White	1.00	800	1.000	1.55	6.24
	Asian minus White	1.00	800	.998	1.02	0.28
	Women minus Men	−0.50	800	1.000	−0.49	0.17
		0.00	800	.998	0.02	0.18
		0.50	800	.995	0.52	0.20
		1.00	800	.993	1.03	0.23
		1.50	800	.993	1.53	0.25
		2.00	800	.994	2.04	0.28
		3.00	800	.995	3.05	0.35
	Low minus High SES	−0.50	800	1.000	−0.49	0.14
		0.00	800	1.000	0.01	0.15
		0.50	800	.998	0.52	0.16
		1.00	800	.998	1.02	0.18
		1.50	800	.998	1.53	0.20
		2.00	800	.998	2.03	0.23
		3.00	800	1.000	3.04	0.27
	Middle minus High SES	−0.50	800	.996	−0.49	0.28
		0.00	800	.991	0.02	0.30

Table S27, continued

Scope	Contrast	g	n	Coverage	Mean ratio	SD of ratio
pooled		0.50	800	.988	0.53	0.33
		1.00	800	.988	1.04	0.37
		1.50	800	.986	1.55	0.42
		2.00	800	.989	2.06	0.47
		3.00	800	.991	3.08	0.57
		1.00	800	.988	1.05	0.38
	Black minus White	−0.50	200	1.000	−0.44	0.46
		0.00	200	.990	0.10	0.57
		0.50	200	.995	0.65	0.73
		1.00	200	.995	1.19	0.92
		1.50	200	.995	1.74	1.12
		2.00	200	.995	2.28	1.33
		3.00	200	1.000	3.37	1.77
		1.00	200	.990	1.55	5.23
		1.00	200	.990	1.02	0.22
		−0.50	200	1.000	−0.49	0.09
		0.00	200	.975	0.02	0.11
		0.50	200	.965	0.52	0.14
		1.00	200	.975	1.03	0.17
		1.50	200	.990	1.53	0.21
		2.00	200	.990	2.04	0.24
		3.00	200	.990	3.05	0.32
	Low minus High SES	−0.50	200	1.000	−0.49	0.08
		0.00	200	1.000	0.01	0.09
		0.50	200	.990	0.52	0.11
		1.00	200	.995	1.02	0.14
		1.50	200	.995	1.53	0.16
		2.00	200	.995	2.03	0.19
		3.00	200	.995	3.04	0.25
		−0.50	200	1.000	−0.49	0.17
	Middle minus High SES	0.00	200	.975	0.02	0.20
		0.50	200	.975	0.53	0.24
		1.00	200	.980	1.04	0.30
		1.50	200	.990	1.55	0.35
		2.00	200	.990	2.06	0.41
		3.00	200	.990	3.08	0.53
		1.00	200	.965	1.05	0.30
		1.00	200	.965	1.05	0.30

Note. Coverage = share of replicates whose interval contains the planted ratio; nominal .986.

Table S28

Controls, Level 2 at the Corpus's Own Precision: Parametric Power

Contrast	g	True region	Conditional, smallest	Conditional, largest	Pairs, smallest	Pairs, largest
Black minus White	−0.50	reversed	4.9 (63.3)	2.8 (33.7)	1.1 (23.6)	0.4 (3.1)
	0.00	missing	0.0 (8.9)	0.0 (3.8)	0.0 (1.7)	0.0 (0.5)
	0.50	attenuated	0.0 (2.0)	0.0 (0.1)	0.0 (0.0)	0.0 (0.0)
	1.00	kept	0.0 (1.0)	0.0 (0.9)	0.0 (0.5)	0.0 (0.3)
	1.50	steepened	0.1 (26.8)	0.2 (15.3)	0.1 (11.6)	0.2 (2.1)
	2.00	steepened	6.5 (70.8)	5.3 (53.7)	4.5 (48.9)	1.6 (8.8)

Table S28, continued

Contrast	<i>g</i>	True region	Conditional, smallest	Conditional, largest	Pairs, smallest	Pairs, largest
Hispanic minus White	3.00	steepened	72.2 (79.0)	62.6 (78.3)	61.6 (78.0)	19.5 (45.9)
	−0.50	reversed	0.5 (18.4)	0.5 (8.0)	0.2 (14.9)	0.1 (0.9)
	0.00	missing	0.0 (2.6)	0.0 (1.4)	0.0 (1.6)	0.0 (0.2)
	0.50	attenuated	0.0 (0.2)	0.0 (0.0)	0.0 (0.1)	0.0 (0.0)
	1.00	kept	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)
	1.50	steepened	0.0 (0.7)	0.0 (0.8)	0.0 (0.5)	0.0 (0.2)
	2.00	steepened	0.0 (8.8)	0.1 (4.8)	0.0 (7.8)	0.1 (1.0)
	3.00	steepened	5.1 (35.9)	3.8 (28.5)	5.1 (35.2)	1.3 (7.4)
Asian minus White	−0.50	reversed	89.3 (100.0)	51.6 (100.0)	24.1 (100.0)	0.7 (3.9)
	0.00	missing	78.9 (100.0)	9.0 (93.4)	0.0 (48.0)	0.0 (0.5)
	0.50	attenuated	44.4 (99.8)	1.3 (82.5)	0.0 (42.0)	0.0 (0.0)
	1.00	kept	0.0 (80.1)	0.0 (53.7)	0.0 (35.7)	0.0 (0.6)
	1.50	steepened	25.9 (100.0)	19.8 (100.0)	14.1 (99.8)	0.7 (4.0)
	2.00	steepened	99.8 (100.0)	99.0 (100.0)	97.3 (100.0)	4.1 (16.1)
	3.00	steepened	100.0 (100.0)	100.0 (100.0)	100.0 (100.0)	39.7 (69.5)
Women minus Men	−0.50	reversed	98.5 (100.0)	75.9 (100.0)	81.5 (100.0)	9.6 (91.9)
	0.00	missing	97.0 (100.0)	52.0 (100.0)	63.4 (100.0)	0.0 (12.9)
	0.50	attenuated	76.8 (100.0)	31.5 (99.9)	42.1 (100.0)	0.0 (4.7)
	1.00	kept	19.9 (100.0)	0.5 (87.4)	1.9 (93.4)	0.0 (12.1)
	1.50	steepened	41.3 (100.0)	32.5 (100.0)	35.1 (100.0)	8.5 (90.2)
	2.00	steepened	100.0 (100.0)	100.0 (100.0)	100.0 (100.0)	85.2 (100.0)
	3.00	steepened	100.0 (100.0)	100.0 (100.0)	100.0 (100.0)	100.0 (100.0)
Low minus High SES	−0.50	reversed	99.7 (100.0)	90.1 (100.0)	62.6 (100.0)	1.7 (20.9)
	0.00	missing	99.2 (100.0)	81.0 (100.0)	25.2 (99.4)	0.0 (1.8)
	0.50	attenuated	86.0 (100.0)	61.3 (100.0)	14.1 (98.6)	0.0 (0.0)
	1.00	kept	37.4 (100.0)	12.6 (99.6)	0.2 (83.2)	0.0 (1.7)
	1.50	steepened	50.6 (100.0)	41.6 (100.0)	32.7 (100.0)	1.8 (20.3)
	2.00	steepened	100.0 (100.0)	100.0 (100.0)	100.0 (100.0)	20.4 (67.4)
	3.00	steepened	100.0 (100.0)	100.0 (100.0)	100.0 (100.0)	95.7 (99.8)
Middle minus High SES	−0.50	reversed	78.8 (100.0)	31.8 (100.0)	20.0 (99.8)	2.7 (41.6)
	0.00	missing	57.4 (99.4)	0.3 (61.2)	0.0 (37.3)	0.0 (2.8)
	0.50	attenuated	14.6 (92.8)	0.0 (39.2)	0.0 (24.1)	0.0 (0.0)
	1.00	kept	0.0 (47.0)	0.0 (28.9)	0.0 (24.2)	0.0 (2.8)
	1.50	steepened	13.4 (100.0)	10.3 (98.9)	9.4 (97.9)	3.0 (39.7)
	2.00	steepened	95.4 (100.0)	87.3 (100.0)	84.9 (100.0)	34.7 (91.4)
	3.00	steepened	100.0 (100.0)	100.0 (100.0)	100.0 (100.0)	99.6 (100.0)
Low minus Middle SES	−0.50	reversed	77.1 (100.0)	33.5 (100.0)	40.3 (100.0)	0.9 (4.8)
	0.00	missing	55.2 (99.7)	0.3 (65.7)	3.1 (78.2)	0.0 (0.8)
	0.50	attenuated	16.2 (95.4)	0.0 (46.3)	0.1 (60.2)	0.0 (0.0)
	1.00	kept	0.0 (52.9)	0.0 (32.3)	0.0 (38.3)	0.0 (0.7)
	1.50	steepened	16.0 (100.0)	11.7 (99.4)	13.4 (99.9)	0.9 (4.5)
	2.00	steepened	97.4 (100.0)	91.5 (100.0)	95.0 (100.0)	4.9 (17.1)
	3.00	steepened	100.0 (100.0)	100.0 (100.0)	100.0 (100.0)	41.3 (70.0)

Note. Each cell: percent of draws naming the true region alone, and in parentheses the percent naming a verdict that contains it. Columns: the simulated-gap standard error, set to the smallest or largest of the four real models under the draw-only (conditional) or the pairs standard error. The largest wrong-verdict rate in any cell is 0.2%.

Table S29

Controls, Level 3: Share of Replicates Passing All Ten Groups

Scope	Shift <i>d</i>	<i>n</i>	0.2 SD	1 point	2 points
per model	0.00	800	84.0 [81.3, 86.4]	98.1 [96.9, 98.9]	100.0 [99.5, 100.0]

Table S29, continued

Scope	Shift d	n	0.2 SD	1 point	2 points
	0.25	800	37.0 [33.7, 40.4]	82.5 [79.7, 85.0]	100.0 [99.5, 100.0]
	0.50	800	1.9 [1.1, 3.1]	27.8 [24.8, 31.0]	100.0 [99.5, 100.0]
	1.00	800	0.0 [0.0, 0.5]	0.0 [0.0, 0.5]	98.1 [96.9, 98.9]
	1.50	800	0.0 [0.0, 0.5]	0.0 [0.0, 0.5]	27.8 [24.8, 31.0]
	2.00	800	0.0 [0.0, 0.5]	0.0 [0.0, 0.5]	0.0 [0.0, 0.5]
	3.00	800	0.0 [0.0, 0.5]	0.0 [0.0, 0.5]	0.0 [0.0, 0.5]
pooled	0.00	200	100.0 [98.1, 100.0]	100.0 [98.1, 100.0]	100.0 [98.1, 100.0]
	0.25	200	91.5 [86.8, 94.6]	100.0 [98.1, 100.0]	100.0 [98.1, 100.0]
	0.50	200	14.0 [9.9, 19.5]	84.5 [78.8, 88.9]	100.0 [98.1, 100.0]
	1.00	200	0.0 [0.0, 1.9]	0.0 [0.0, 1.9]	100.0 [98.1, 100.0]
	1.50	200	0.0 [0.0, 1.9]	0.0 [0.0, 1.9]	84.5 [78.8, 88.9]
	2.00	200	0.0 [0.0, 1.9]	0.0 [0.0, 1.9]	0.0 [0.0, 1.9]
	3.00	200	0.0 [0.0, 1.9]	0.0 [0.0, 1.9]	0.0 [0.0, 1.9]

Note. Shift d = points added to every persona mean. Shares in percent with 95% intervals. At $d = 0$ the share is the pass rate of real people; at $d > 0$ a failure is the correct reading.

Table S30

Controls, Level 4: Share of Replicates Passing Each Rule

Condition	Dose (%)	Unit	n	R1	R2	R3	Pass	λ_1/λ_2	ϕ	Within λ_1/λ_2
L4 decorrelate	25.0	clinical	200	100.0	100.0	100.0	–	5.68	.999	3.46
L4 decorrelate	25.0	model	200	100.0	100.0	100.0	100.0	–	–	–
L4 decorrelate	25.0	narrative	200	100.0	100.0	100.0	–	5.66	.999	3.46
L4 decorrelate	50.0	clinical	200	100.0	100.0	0.0	–	4.13	.999	2.55
L4 decorrelate	50.0	model	200	100.0	100.0	0.0	0.0	–	–	–
L4 decorrelate	50.0	narrative	200	100.0	100.0	0.0	–	4.13	.999	2.56
L4 decorrelate	75.0	clinical	200	7.5	100.0	0.0	–	2.75	.997	1.71
L4 decorrelate	75.0	model	200	1.5	100.0	0.0	0.0	–	–	–
L4 decorrelate	75.0	narrative	200	8.5	100.0	0.0	–	2.78	.997	1.72
L4 decorrelate	100.0	clinical	200	0.0	20.0	0.0	–	1.50	.982	1.03
L4 decorrelate	100.0	model	200	0.0	8.5	0.0	0.0	–	–	–
L4 decorrelate	100.0	narrative	200	0.0	24.5	0.0	–	1.51	.982	1.03
Positive control	0.0	clinical	200	100.0	100.0	100.0	–	7.56	.999	4.44
Positive control	0.0	model	200	100.0	100.0	100.0	100.0	–	–	–
Positive control	0.0	narrative	200	100.0	100.0	100.0	–	7.53	.999	4.45
reference half	0.0	reference	200	100.0	–	100.0	–	–	–	4.25

Note. Dose = share of draws rebuilt item by item from other draws of the same persona. R1, R2, R3 and Pass are shares of replicates in percent; Pass is scored on R1, R2 and R3 in both framings (unit `model`). R4 was not simulated. The last three columns are means over replicates. The reference-half row reads the NHANES half used as reference.

S6. External Level 3: An Illustration

This section is an illustration of level 3 on two external releases, and it is not a verdict on either. Level 3 needs a use tolerance that the deployer states, and neither group of original authors stated one. The use tolerances below are placeholders set by the present analyst. On OpinionQA a group passes when the model’s plurality answer matches the Pew plurality answer on at least 90% of questions. On Argyle et al. Study 3 the use

tolerance is 5 percentage points for proportions and one response category for ordinal scales. The strict tolerance is the human group mean’s own 95% sampling half-width.

OpinionQA.. The steered model distributions and the Pew American Trends Panel targets come from the Meister et al. release. The Pew targets in that release are unweighted counts, so the comparison is to the unweighted Pew sample. Of 268 method, model, steering and group cells, 0 pass the placeholder use tolerance (Table S31). The models match the Pew plurality answer on 25.5% to 79.7% of questions (median 57.1%), and a fresh Pew sample of the same size would match it on a median of 94.5%.

Argyle et al. Study 3. GPT-3 answered for the same American National Election Studies respondents whose answers form the reference, and the ANES file carries no weights. The released listing for Study 3 sets `temperature=0.001` inside the generation loop, while the release’s README calls the main file the 0.7 run. This analysis follows the README and labels that run `t0.7_main`. In the main run, 4 of 116 outcome and group cells pass the strict tolerance and 57 pass the placeholder use tolerance (Tables S32 and S33). The receipts are `paper_brm/external/results/opinionqa/level3_groups.csv` and `paper_brm/external/results/argyle/level3_groups.csv`, and the method is described in `paper_brm/external/RESULTS.md`.

Table S31

External Level 3 Illustration: OpinionQA Steered Distributions Against Pew

Method	Model	Steering	Groups	Plurality match, median (range)	Human resample	Within strict	Mean R	Groups passing
express_ distribution	anthropic_haiku	few-shot	14	57.1 (47.9–66.3)	94.3	16.8	0.10	0
express_ distribution	anthropic_haiku	persona	14	60.2 (36.6–62.7)	94.3	13.8	0.12	0
express_ distribution	anthropic_opus	few-shot	14	66.3 (61.8–78.6)	94.3	18.5	0.08	0
express_ distribution	anthropic_opus	persona	14	67.0 (45.6–79.7)	94.5	20.3	0.09	0
express_ distribution	gpt-3.5-turbo-0125	few-shot	14	46.2 (39.8–52.0)	94.3	10.3	0.16	0
express_ distribution	gpt-3.5-turbo-0125	persona	14	46.6 (26.5–51.6)	94.7	12.8	0.13	0
express_ distribution	gpt-4	few-shot	14	60.5 (54.1–68.4)	94.3	18.9	0.09	0

Table S31, continued

Method	Model	Steering	Groups	Plurality match, median (range)	Human resample	Within strict	Mean R	Groups passing
express_ distribution	gpt-4	persona	14	63.0 (52.8–74.5)	94.3	15.3	0.09	0
express_ distribution	llama3-70b	few-shot	6	58.2 (56.1–66.3)	96.2	14.3	0.11	0
express_ distribution	llama3-70b	persona	6	57.7 (50.0–63.3)	96.2	11.7	0.11	0
model_logprobs	gpt-3.5-turbo-0125	few-shot	14	46.0 (40.2–50.0)	94.3	5.1	0.23	0
model_logprobs	gpt-3.5-turbo-0125	persona	14	45.4 (39.8–54.1)	94.3	12.3	0.16	0
model_logprobs	gpt-4	few-shot	14	66.0 (57.7–73.5)	94.3	6.7	0.20	0
model_logprobs	gpt-4	persona	14	60.5 (43.7–69.4)	94.3	4.6	0.24	0
model_logprobs	llama3-70b	few-shot	6	60.2 (54.6–63.3)	95.3	6.1	0.19	0
model_logprobs	llama3-70b	persona	6	62.2 (52.0–69.4)	95.3	3.6	0.22	0
sequence	anthropic_haiku	few-shot	8	54.1 (47.4–59.2)	94.3	11.3	0.12	0
sequence	anthropic_haiku	persona	8	49.0 (36.7–55.1)	94.3	11.2	0.13	0
sequence	anthropic_opus	few-shot	8	64.8 (58.2–74.5)	94.3	10.7	0.12	0
sequence	anthropic_opus	persona	8	58.2 (48.0–66.3)	94.3	11.3	0.13	0
sequence	gpt-3.5-turbo-0125	few-shot	8	45.4 (41.8–50.0)	94.3	11.3	0.16	0
sequence	gpt-3.5-turbo-0125	persona	8	34.2 (25.5–40.8)	94.3	6.7	0.16	0
sequence	gpt-4	few-shot	8	60.2 (56.7–68.4)	94.3	15.8	0.10	0
sequence	gpt-4	persona	8	55.1 (49.5–60.2)	94.3	13.8	0.11	0
sequence	llama3-70b	few-shot	6	58.2 (51.0–66.3)	95.3	9.7	0.16	0
sequence	llama3-70b	persona	6	58.7 (56.7–63.3)	95.3	7.1	0.16	0

Note. Shares in percent, medians over groups. Plurality match = share of questions on which the model’s plurality answer equals the Pew plurality answer. Human resample = the same share for a fresh Pew sample of the same size. Within strict = share of questions whose residual lies inside the human group’s own 95% sampling half-width. Mean |R| = mean absolute residual as a share of the scale range. Groups passing = groups whose plurality match is at least 90%, the placeholder use tolerance.

Table S32

External Level 3 Illustration: Argyle et al. Study 3, Passing Cells by Run

Run	Outcome × group cells	Pass strict	Pass use
t0.001	116	8	51
t0.7_main	116	4	57
t1.0	116	1	58

Note. Strict tolerance: the human group mean’s own 95% sampling half-width. Use tolerance: a placeholder of 5 percentage points for proportions and one response category for ordinal scales.

Table S33

External Level 3 Illustration: Argyle et al. Study 3, Main Run by Outcome

Outcome	All respondents						Groups				
	Human	Silicon	Residual	90% CI	Use tol.	Pass	<i>n</i>	Strict	Use	Agree (%)	
Trump share among voters	0.46	0.26	−0.20	[−0.22, −0.18]	0.05	No	13	0	2	72.3	
Trump two-party share	0.52	0.50	−0.02	[−0.04, −0.01]	0.05	Yes	12	3	4	86.2	
Turnout 2016	0.87	0.83	−0.04	[−0.05, −0.03]	0.05	No	13	1	1	80.3	
Attends church	0.60	0.60	0.00	[−0.02, 0.02]	0.05	Yes	13	0	1	52.4	
Ever discusses politics	0.82	0.88	0.06	[0.05, 0.08]	0.05	No	13	0	1	76.9	

Table S33, continued

Outcome	All respondents						Groups			
	Human	Silicon	Residual	90% CI	Use tol.	Pass	<i>n</i>	Strict	Use	Agree (%)
Party ID (1-7)	3.85	4.39	0.54	[0.49, 0.60]	1.00	Yes	13	0	11	23.6
Ideology (1-7)	4.16	4.04	-0.12	[-0.17, -0.07]	1.00	Yes	13	0	13	22.0
Patriotism (1-7, 1 = flag feels extremely good)	1.95	1.43	-0.52	[-0.57, -0.48]	1.00	Yes	13	0	11	44.6
Political interest (1-4, 1 = very)	2.14	1.74	-0.40	[-0.43, -0.37]	1.00	Yes	13	0	13	32.6

Note. Main run (`t0.7_main`). Proportions for the first five outcomes, scale points for the rest. Groups: all respondents plus twelve age, education, gender, party and race groups. Agree = share of respondents whose silicon answer equals their own answer.